

UNIVERSITY OF CALIFORNIA
Los Angeles

Vowel System Typology
and Confusion Minimization

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by

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ABSTRACT OF THE THESIS

Vowel System Typology and Confusion Minimization

by

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Using a novel algorithm called Iterated Confusion Minimization, this work unifies diachronic and synchronic approaches to vowel system typology. Compatible with sound change models that involve confusion-reducing adjustments to phonetic distributions (Blevins, 2004, p. 285-289; Boersma and Hamann, 2008), Iterated Confusion Minimization simulates the interaction of optimal listeners and speakers over time. Simulations over random initial vowel systems resemble observed sound change (e.g. incremental chain shifts) and most often result in typologically dominant systems with minimal confusion, suggesting a diachronic explanation for the typological dominance of perceptually optimal vowel systems (cf. Liljencrants and Lindblom, 1972). In contrast to past vowel dispersion models (e.g. Schwartz et al., 1997b), Iterated Confusion Minimization easily accounts for interior vowels like /ə/, allowing for prediction of the most common system up to 11 vowels. Finally, variation in simulation output mirrors typological variation, with a tight correlation between output frequency and typological frequency up to 6 vowels that outperforms the baseline set by de Boer (1999).

The thesis of John Francis McGahay is approved.

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CHAPTER 1

Overview

The observation that natural languages tend to make use of dispersed vowel systems with high degree of perceptual contrast between categories has a long history in phonetic theory. The last century has produced a large body of work investigating vowel dispersion from various angles, ranging from analyses of typological databases that establish the basic facts of how vowel systems pattern across languages (Sedlak, 1969; Crothers, 1978; Schwartz et al., 1997a; Becker-Kristal, 2010), to synchronic¹ models collectively called Dispersion Theory that characterize typologically dominant vowel systems as perceptually optimal (Liljencrants and Lindblom, 1972; Lindblom, 1986; Schwartz et al., 1997b; Flemming, 1995), to diachronic models that explain how phoneme dispersion might emerge through sound change (de Boer, 1999; Blevins, 2004, p. 285-289; Boersma and Hamann, 2008).

Despite considerable success, there does not yet exist any truly comprehensive model of vowel typology capable of addressing these various aspects of dispersion simultaneously. While models of optimal vowel systems within Dispersion Theory may give a phonetic substance-based account for the dominant systems across the world's languages, such models tend to lack an explanation of how these optimal systems emerge and often fail to explain typological variation (Liljencrants and Lindblom, 1972; Lindblom, 1986; Schwartz et al., 1997b). Where a model has predicted the emergence of such typological variation at realis-

¹This work labels as 'synchronic' any model that evaluates sound systems based on intrinsic characteristics, in contrast to overtly diachronic models of sound system emergence. Some authors of 'synchronic' models may attribute dispersion pressures to diachrony (e.g. Lindblom, 1986) rather than the synchronic grammar (e.g. Flemming, 1995).

tic rates, the sound change mechanism involved has proved unrealistic (de Boer, 1999, 2000). Meanwhile, accounts of emergent dispersion offering plausible models of sound change have tended to explore cases more limited in scope than accounting for typological frequency of competing vowel systems (Blevins, 2004, p. 285-289; Boersma and Hamann, 2008).

The current work seeks to fill this gap by unifying these various facets of vowel system typology into a single comprehensive model, achieved through use of a novel algorithm called Iterated Confusion Minimization (ICM).

ICM models the interaction of mathematically optimal listeners and speakers, explored in Chapter 2. This interaction is shaped by a perception-production asymmetry, referred to here as the *confusion paradox*. As we will see in Section 2.2, given some system of phonetic distributions and a listener following maximum likelihood behavior (Fisher, 1922) optimized for that system, the mathematically optimal phonetic system a speaker can adopt to minimize that listener’s confusion is *not* in general the original system for which the listener is optimized. Whereas one might naïvely suppose that a speaker will minimize an optimal listener’s confusion by converging to the exact phonetic system of that listener, it is demonstrated that the speaker can typically do better by adopting a system close to the listener’s system but with slightly greater dispersion, *even if the listener is expecting to hear their own system*.

Assuming that individuals follow roughly optimal speaker behavior and tend to innovate sound systems in this way, we should expect listeners to accommodate their interlocutors by re-optimizing their perceptual behavior against these new phonetic distributions (cf. Clarke and Garrett, 2004). However, these new systems are susceptible to the same confusion paradox that motivated speakers to adopt a new system in the first place, meaning that optimal speakers should adopt newer, even more dispersed systems, in response to which optimal listeners should re-optimize their perception yet again, and so on and so forth.

The ICM algorithm given in Section 2.3 models this resulting game of perceptual cat and mouse between optimal listeners and speakers who iteratively adjust their phonetic

behavior in response to one another. ICM simulations result in incremental, dispersion-increasing change to an initial sound system over time until an equilibrium system with high dispersion is attained. Such equilibrium systems are stable under ICM because they are perceptually *self-optimal*, i.e. an optimal speaker can no longer reduce the confusion of a listener optimized for such a system by ‘gaming’ that listener’s perception with a new system of phonetic distributions, thus overcoming the asymmetry of the confusion paradox.

The basic mechanism of sound change under ICM, i.e. restructuring of phonetic distributions to include a higher proportion of less easily confused tokens, is broadly compatible with existing diachronic models put forward by Blevins (2004, p. 285-289) to account for chain shifts using Exemplar Theory (Pierrehumbert, 2001) and Boersma and Hamann (2008) to account for patterns of sibilant dispersion using Stochastic Optimality Theory (Boersma et al., 1997; Boersma and Hayes, 2001). However, by abstracting away from the implementational details of specific phonological frameworks into more general mathematical terms, ICM makes much more explicit the link between a plausible model of sound change and a quantitative measure of perceptual contrast, i.e. maximum likelihood confusion (see Section 2.1).

Mathematically guaranteed to only ever improve the maximum likelihood confusion of a sound system, ICM and the sound change it models can be understood as a local optimization algorithm with respect to confusion. Since local optimization algorithms often provide an efficient way to discover globally optimal solutions, ICM thus suggests a diachronic explanation for the typological dominance of vowel systems with maximal perceptual contrast, successfully captured by past work in Dispersion Theory, albeit without a satisfactory explanatory mechanism.

Using ICM to explain vowel system typology requires a number of modeling decisions for implementation over the vowel space, discussed in Chapter 3. Chapter 4 explores some example simulations, which reveal that ICM over random initial vowel systems, even extremely unnatural systems, quickly evolve into stable, typologically attested systems with

low confusion and high dispersion.

Vowel system evolution under ICM follows naturalistic trajectories reminiscent of observed sound change, with incremental change over time, frequently in the form of chain shifts along the periphery of the vowel space, as illustrated in Section 4.1. Given the ubiquity of such changes in the phonological histories of the world’s languages (cf. Labov, 1994, p. 113-291), ICM thus provides a robust, principled model of a significant aspect of natural language sound change.

Past work in Dispersion Theory has generally evaluated perceptual contrast using so-called ‘energy functions’ defined based on the psychoacoustic distance between vowel categories (e.g., Liljencrants and Lindblom, 1972; Schwartz et al., 1997b). However, by directly modeling listener confusion rather than using energy functions merely correlated with confusion, the current approach to evaluating perceptual contrast is significantly less idealized. Section 4.2 illustrates how this reduction in idealization allows for the straightforward prediction of vowel systems with interior vowels like /ə/, the non-prediction of which has proved a pervasive limitation of past dispersion models by contrast.

Chapter 5 systematically evaluates the results of 62,000 ICM simulations over randomly initialized vowel systems under different parameter settings. Section 5.1 shows that ICM simulations most often result in typologically dominant vowel systems with optimal maximum likelihood confusion. Whereas the influential dispersion model of Schwartz et al. (1997b) only predicts typologically dominant systems up to 7 vowels, one ICM parameterization predicts every single dominant system up to 11 vowels. This improvement appears owing to the aforementioned ability of ICM to predict interior vowels common in larger vowel inventories.

Beyond these typologically dominant systems at global optima with respect to confusion, a variety of further vowel systems at local optima are also stable under ICM, generally corresponding to attested (albeit typologically non-dominant) systems. This allows for comparison of ICM emergence rate of different vowel systems against their observed typological frequency, analyzed in Section 5.2.

The best ICM parameterization predicts every vowel system up to 7 vowels attested in at least two primary vowel systems in the UCLA Phonological Segment Inventory Database (Maddieson, 1984). For inventory sizes up to 6, this parameter settings shows high correlation with typological frequency, with near-categorical prediction of /iau/ and /ieaou/ for 3- and 5-vowel systems and variation for 4- and 6-vowel system outputs matching typological variation. Using Kullback-Leibler divergence (Kullback and Leibler, 1951) to evaluate ICM emergence rate of different systems, ICM generally provides a better fit to typological frequency than the baseline set by de Boer (1999). Although correlation with typological frequency worsens for inventory sizes 7 and larger, some possible explanations for this correlation breakdown are put forward which might be investigated in future work.

CHAPTER 2

Iterated Confusion Minimization

Although previous work has made significant strides in describing and explaining vowel dispersion from both synchronic and diachronic angles, a unified account of vowel system typology has not yet been established.

Some desiderata for a comprehensive account combining achievements of past work are as follows:

1. A quantitative measure of perceptual contrast which evaluates the typologically most frequent vowel systems as optimal (cf. Liljencrants and Lindblom, 1972; Schwartz et al., 1997b, etc.)
2. The ability to predict emergence of naturalistic vowel systems at rates comparable to typological frequency (cf. de Boer, 1999, 2000)
3. A model of emergent dispersion consistent with observed sound change (cf. Blevins, 2004, p.285-289; Boersma and Hamann, 2008)

The key to satisfying these desiderata in the current work lies in a novel algorithm and model of sound change called Iterated Confusion Minimization (ICM), which captures the diachronic interaction of mathematically optimal listeners and speakers investigated in the next sections.

2.1 Optimal Listener Behavior

Assuming that all phoneme categories have equal probability, an optimal listener follows the perceptual behavior of a maximum likelihood classifier (Fisher, 1922).¹ As we will see, maximum likelihood classifiers appealingly provide a mathematically precise definition of confusion for well-defined phonetic distributions.

Given a system of phonemes with associated phonetic distributions $S = \{\phi_1, \phi_2, \dots\}$, a maximum likelihood classifier C optimized for S categorizes a phonetic token x as whichever phoneme ϕ in S is most likely to produce a token with the value of x . This is expressed formally in Equation 2.1:²

$$C(x) = \underset{\phi \in S}{\operatorname{argmax}} P[\phi = x] \quad (2.1)$$

Figure 2.1 illustrates this behavior with a schematic example of two phonemes that take overlapping distributions along some one-dimensional phonetic continuum. A maximum likelihood classifier always perceives a token along this continuum as whichever phoneme category gives that token the highest likelihood. This is equivalent to whichever phoneme has a higher probability density function evaluated at the phonetic value of that token, represented in Figure 2.1 by the height of the bell curve associated with each phoneme.

In Figure 2.1, we can see that Phoneme 1 is more likely than Phoneme 2 for all tokens with phonetic value less than 1 and Phoneme 2 is more likely than Phoneme 1 for all tokens with phonetic value greater than 1. Hence, a maximum likelihood classifier will categorically perceive tokens at phonetic values less than 1 as Phoneme 1 and tokens at phonetic values

¹An optimal listener actually follows principles of Bayesian inference (Gelman et al., 1995), although maximum likelihood and Bayesian behavior are equivalent under the assumption of equiprobable phonemes. Modeling Bayesian classifiers involves significantly more complicated treatment of prior and conditional probabilities left to future work.

²An abuse of notation is made in Equation 2.1 and elsewhere for simplicity. A maximum likelihood classifier over a continuous space maximizes not probability $P[\phi = x]$ (typically 0 for any continuous random variable ϕ and point x), but probability *density*. Since some discretization is always required in practice, this distinction is largely academic.

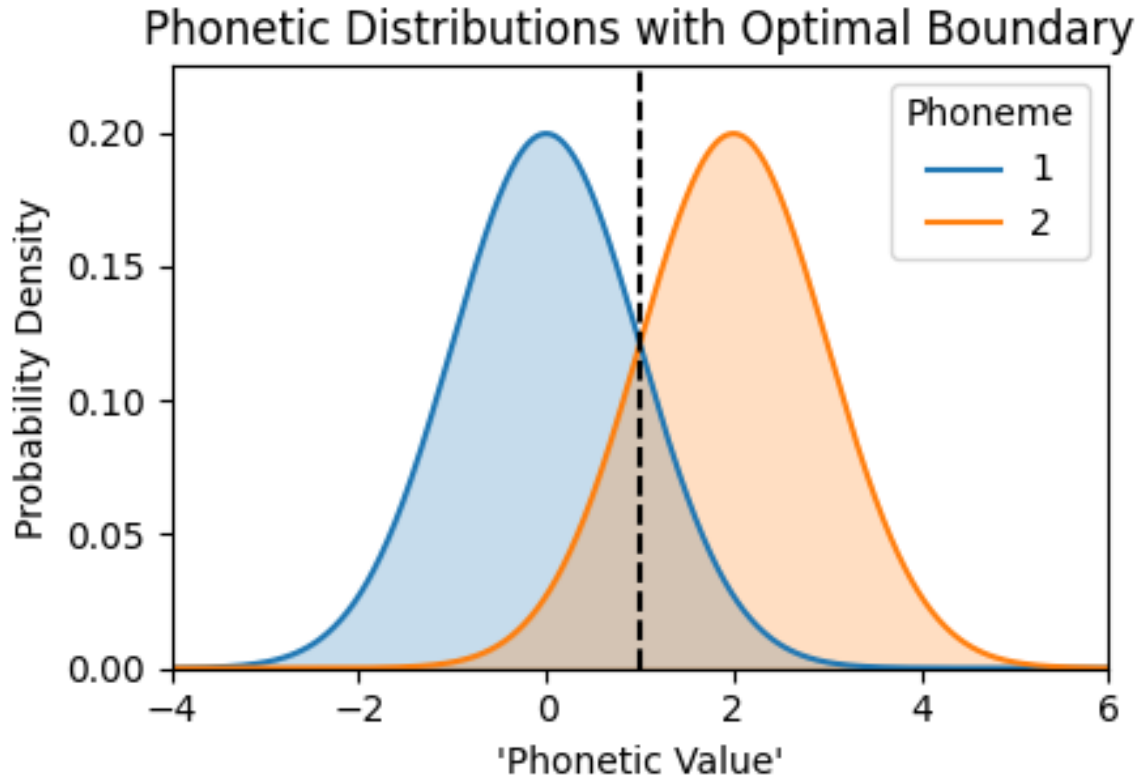


Figure 2.1: Two idealized phonetic distributions. The optimal perceptual boundary lies at the intersection point of these bell curves, i.e. the ‘phonetic value’ of 1 at which these ‘phonemes’ are equiprobable. The confusion rate of a maximum likelihood classifier over this pair is the area under each curve on the ‘wrong’ side of this boundary, equal to the area of the triangular region of overlap between distributions, in this case 15.87%.

greater than 1 as Phoneme 2. Based on this behavior, a maximum likelihood classifier can be described as having a sharp perceptual boundary at the points of intersection of the probability density functions of the phonemes over which it is defined, represented in Figure 2.1 with a dashed vertical line.

The confusion rate $\text{conf}(C|S)$ of a classifier C given a sound system S (whether or not C is optimized for S) has the following formal definition:^{3,4}

$$\begin{aligned}\text{conf}(C|S) &= \frac{1}{|S|} \sum_{\phi \in S} P[C(\phi) \neq \phi] \\ &= \frac{1}{|S|} \sum_{\phi \in S} \int P[\phi = x] P[C(x) \neq \phi] dx\end{aligned}\tag{2.2}$$

This is simply the normalized sum of the phoneme-level confusion rate $P[C(\phi) \neq \phi]$, i.e. probability C misidentifies ϕ , for each phoneme ϕ in S , which is calculated through integration over the phonetic space.

For classifiers with sharp categorical perceptual boundaries such as maximum likelihood classifiers, the definition of confusion given in Equation 2.2 has a much more intuitive definition: the percentage of tokens on the ‘wrong side’ of each perceptual boundary. In Figure 1, this corresponds to the area under the blue curve to the right of the boundary plus the area under the orange curve to the left of the boundary, which comes out to 0.1587.⁵

Although Equation 2.2 gives the general confusion rate $\text{conf}(C|S)$ given a sound system S and arbitrary classifier C not necessarily optimized for S , the current work often makes use (e.g., in the last paragraph with respect to Figure 2.1) of a maximum likelihood confusion rate $\text{conf}_{ml}(S) = \text{conf}(C_S|S)$ given S and a non-arbitrary maximum likelihood classifier C_S optimized on S . This value has a special role in the current work as a system-intrinsic

³To generalize Equation 2.2 away from the assumption of equiprobable phonemes, replace the normalizing constant $1/|S|$ with a weight $P[\phi]$ on each term of the summation.

⁴The integral generalizes to higher dimensions, i.e. $\int_{\mathbb{R}^n} P[\phi = \mathbf{x}] P[C(\mathbf{x}) \neq \phi] d\mathbf{x}$

⁵Confusion is exactly equal to this area in Figure 2.1 only because the total area under the two bell curves has been normalized to 1 (double-counting the overlap region).

quantitative measure of perceptual contrast, providing an alternative to energy functions used in past dispersion models.⁶ It is given below by substituting in the righthand side of Equation 2.1 for $C(x)$ in Equation 2.2:

$$\begin{aligned} \text{conf}_{ml}(S) &= \frac{1}{|S|} \sum_{\phi \in S} P \left[\left(\underset{\psi \in S}{\operatorname{argmax}} P[\psi = \phi] \right) \neq \phi \right] \\ &= \frac{1}{|S|} \sum_{\phi \in S} \int P[\phi = x] P \left[\left(\underset{\psi \in S}{\operatorname{argmax}} P[\psi = x] \right) \neq \phi \right] dx \end{aligned} \quad (2.3)$$

More intuitively, the maximum likelihood confusion $\text{conf}_{ml}(S)$ of a sound system S is simply the amount of overlap between its phonetic distributions, which corresponds to the triangular overlap area in Figure 2.1.

2.2 Optimal Speaker Behavior

Having characterized the general behavior of the optimal listener, a natural next step is to ask how the optimal speaker behaves when interacting with an optimal listener. In order for this question to be interesting for the linguist, we have to constrain the options of the optimal speaker at least somewhat. Unconstrained, the optimal speaker could simply adopt point-like phonetic distributions with zero variance on the correct sides of the optimal listener’s perceptual boundaries for each category. This would yield zero phonetic overlap between categories and correspondingly zero confusion on the part of the listener. However, it is clear that natural language phonemes take highly variable phonetic distributions with heavy overlap, so such behavior is unrealistic for the optimal human speaker.

Achieving rigor in identifying the set of possible phonetic distributions available to a speaker is non-trivial, and this issue will be addressed with respect to the vowel space in Section 3.2. For the purpose of illustration in the current section, however, we will

⁶For a system of n vowels, energy is defined as $\sum_{i=1}^{n-1} \sum_{j=i+1}^n 1/d_{ij}^2$ where d_{ij} is the pairwise Euclidean distance between the i th and j th vowel, typically measured in a psychoacoustic scale. Perceptual contrast is considered maximized when energy is minimized, which occurs when all pairwise distances between vowels are maximized.

assume that all phonetic distributions follow one-dimensional normal distributions with fixed variance.

Figure 2.2 shows a sound system of three such phonetic distributions, represented by the three bell curves with solid outlines. Let us suppose that some individual, Speaker A, has these distributions and is also an optimal listener for this (i.e. their own) system, with maximum likelihood perceptual boundaries at the points of overlap marked with the black vertical lines. If another individual, Speaker B, wants to minimize the confusion Speaker A experiences as an optimal listener when Speaker B talks, what phonetic distributions should Speaker B adopt as an optimal speaker? Although one might intuitively believe that that Speaker B should simply adopt the exact same system as Speaker A, this intuition turn out false upon closer investigation.

Consider the middle distribution of Figure 2.2 in orange with the solid outline. We can see that this distribution has very high confusion to its left (given the large overlap area to the left of the perceptual boundary in black), but virtually no confusion to its right.

Consider now the alternative middle distribution shown with a dotted outline. Although this distribution now has slightly greater overlap with the distribution to the right, we can see that the area beneath this bell curve on the ‘wrong’ side of the optimal perceptual boundaries is much smaller than for the original distribution due to decrease in overlap on the problematic left side. This means that using this new dotted distribution will result in lower confusion for Speaker A than will using A’s original distribution, *even though* Speaker A has perceptual boundaries optimized for this original distribution rather than this new distribution. This means Speaker B, if acting as an optimal speaker, should adopt this new distribution to minimize confusion, in effect ‘gaming’ Speaker A’s perceptual boundaries relative to the baseline performance of A’s own original distributions.

This counter-intuitive asymmetry between the optimal perceptual boundaries for a sound system and the optimal sound system for a set of perceptual boundaries can be expressed as the following veridical paradox:

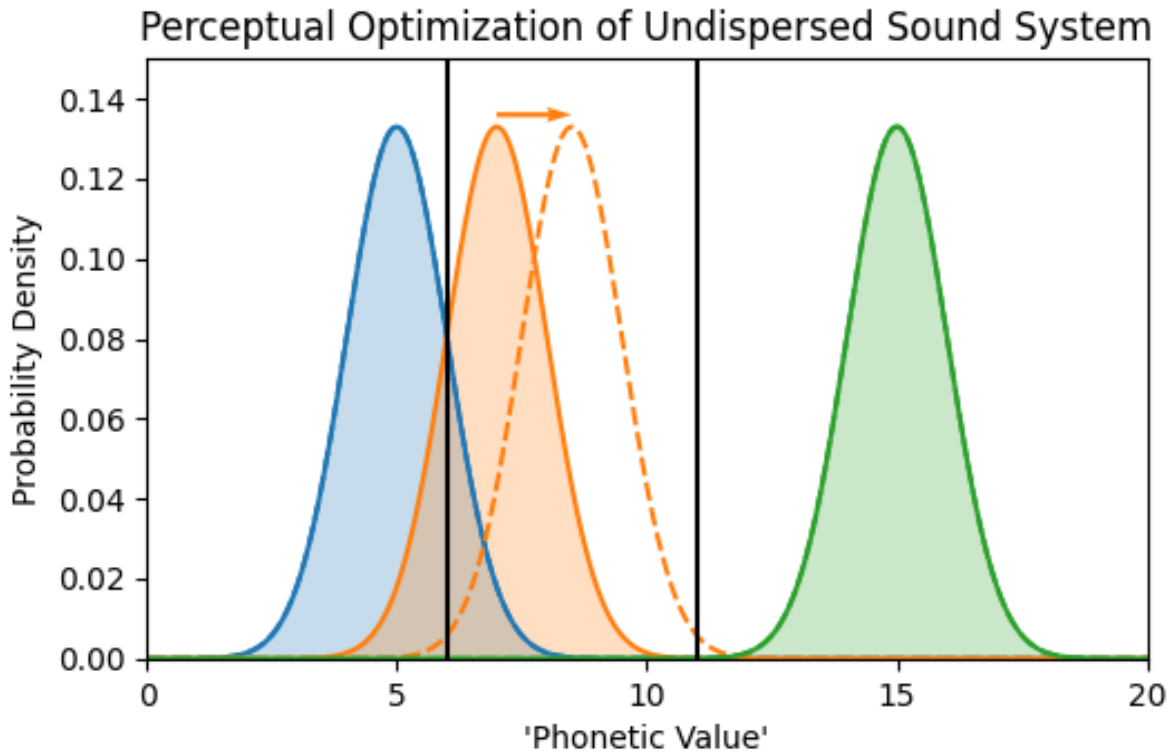


Figure 2.2: Illustration of the confusion paradox. Given the system with solid outlines and a listener with optimal perceptual boundaries (black vertical lines) for that system, an optimal speaker can reduce this optimal listener’s confusion (percent of distribution on ‘wrong’ side of boundaries) by shifting the phoneme to the dashed distribution, even though this is not the system for which the listener’s perception is optimized.

Confusion Paradox: Given a set of phonetic distributions S and a classifier C optimized over S , the optimal set of phonetic distributions S' that minimizes the confusion rate of C is not in general equivalent to the original set of phonetic distributions S for which C is optimized.

The confusion paradox has far-reaching implications for sound change if humans indeed follow (roughly) optimal listener and speaker behavior. We can observe in this example that Speaker B's perceptually optimizing change to the middle distribution shifts this system closer to a maximally dispersed system with even spacing. This immediate result shown by the dashed line in Figure 2.2 represents a local optimization, which still has a ways to go towards the maximally dispersed, globally optimal ideal with even spacing.

However, we will see that when this sort of local optimization takes place repeatedly over time, as modeled by the Iterated Confusion Minimization (ICM) algorithm introduced in the next section, such maximal dispersion can indeed be attained. The confusion paradox thus gives ICM, the heart of the current work, a mathematically principled basis as an engine of sound change towards globally optimal systems rooted in locally optimizing listener and speaker behavior.

Before proceeding, it is helpful to formally define the optimal speaker behavior explored in this section. Given a listener who follows the perceptual behavior of a classifier C over S and a set of possible phonetic distributions Φ , this speaker should adopt a system S' composed of each least confusing phonetic distribution ϕ' in Φ corresponding to a phoneme ϕ in S by minimizing the phoneme-level confusion $P[C(\phi') \neq \phi]$, calculated through integration over the phonetic space. More compactly, the optimal speaker should adopt the system S' defined as follows:

$$\begin{aligned} S' &= \{ \phi' \mid \underset{\phi' \in \Phi}{\operatorname{argmin}} P[C(\phi') \neq \phi], \phi \in S \} \\ &= \{ \phi' \mid \underset{\phi' \in \Phi}{\operatorname{argmin}} \int P[\phi' = x] P[C(x) \neq \phi] dx, \phi \in S \} \end{aligned} \tag{2.4}$$

It is worth noting that this optimization is entirely local from the speaker’s perspective, i.e. to find the least confusing phonetic distribution ϕ' for a given phoneme ϕ in S , the optimal speaker only needs to know the perceptual boundaries of ϕ , without consideration of the phonetic distribution of any other phoneme in S or S' .

2.3 The Iterated Confusion Minimization Algorithm

In the last section, we saw that an optimal speaker can improve an optimal listener’s confusion by shifting their phonetic distributions away from the listener’s distributions, even as the optimal listener maintains fixed perceptual behavior optimized for their own system. However, in reality we should expect an optimal listener to shift their perceptual boundaries in response to changes in an interlocutor’s phonetic distributions. Indeed, such phonetic accommodation is a well-established human perceptual behavior (cf. Clarke and Garrett, 2004; Magnuson and Nusbaum, 2007; Kleinschmidt and Jaeger, 2015).

Once these perceptual boundaries are updated, however, the new system is susceptible to the same confusion paradox as the original system, meaning that an optimal speaker should adjust their distributions yet again, in response to which the optimal listener should update their perception yet again, and so on and so forth.

This interaction of optimal listener and speaker who iteratively update their respective behavior is captured by a new algorithm named Iterated Confusion Minimization (ICM), expressed as pseudocode in Algorithm 1. Lines 4 and 6-9 boxed in blue dashes emulate the behavior of an optimal listener and speaker respectively as described by Equations 2.1 and 2.4 of the previous sections. The previous illustration of the confusion paradox in Figure 2.2 in effect corresponds to a single iteration of this algorithm.

Figure 2.3 shows an ICM simulation over an initially undispersed sound system S composed of four phonemes. After an initial ICM iteration (top left), updates to the maximum likelihood perceptual boundaries allow a second ICM iteration to take place (bottom left).

Algorithm 1 Iterated Confusion Minimization (ICM)

Input: $S_0 \subset \Phi$: an initial sound system S_0 composed of distributions within Φ , where Φ denotes the set of possible phonetic distributions

Output: $S_i \subset \Phi$, s.t. $|S_i| = |S_0|$: the sound system S_i after i iterations with the same number of phonemes as the initial system S_0

```
1:  $i \leftarrow 0$  ▷ start iteration counter
2: repeat ▷ iterate steps below
3:   ▷ Optimal listener behavior
4:    $C_i(x) \leftarrow \operatorname{argmax}_{\phi \in S_i} P[\phi = x]$  ▷ define optimal classifier function
5:   ▷ Optimal speaker behavior
6:    $S_{i+1} \leftarrow \{\}$  ▷ initialize new sound system as empty
7:   for all  $\phi \in S_i$  do ▷ for each preexisting phoneme:
8:      $\phi' \leftarrow \operatorname{argmin}_{\psi \in \Phi} P[C_i(\psi) \neq \phi]$  ▷ find least confusing distribution
9:      $S_{i+1} \leftarrow S_{i+1} \cup \{\phi'\}$  ▷ and add it to new system
10:   $i \leftarrow i + 1$  ▷ update iteration counter
11: until  $S_i = S_{i-1}$  ▷ stop iterating once system is stable
12: return  $S_i$  ▷ output stable sound system
```

This process repeats until a maximally dispersed system with even spacing emerges within 10 ICM iterations (top right).

Even using highly idealized phonetic distributions, this example gives a realistic illustration of a chain shift. As Phoneme 3 shifts to fill the gap to its right, Phoneme 2 incrementally shifts in a pull chain to the original position of Phoneme 3, eventually yielding an equilibrium with even spacing and minimal confusion. The stability of this system in the top right results from its perceptual *self-optimality*. Since each distribution is already maximally far away from its maximum likelihood boundaries, the confusion rate of an optimal listener cannot be improved through a speaker’s phonetic adjustments, thus overcoming the asymmetry of the

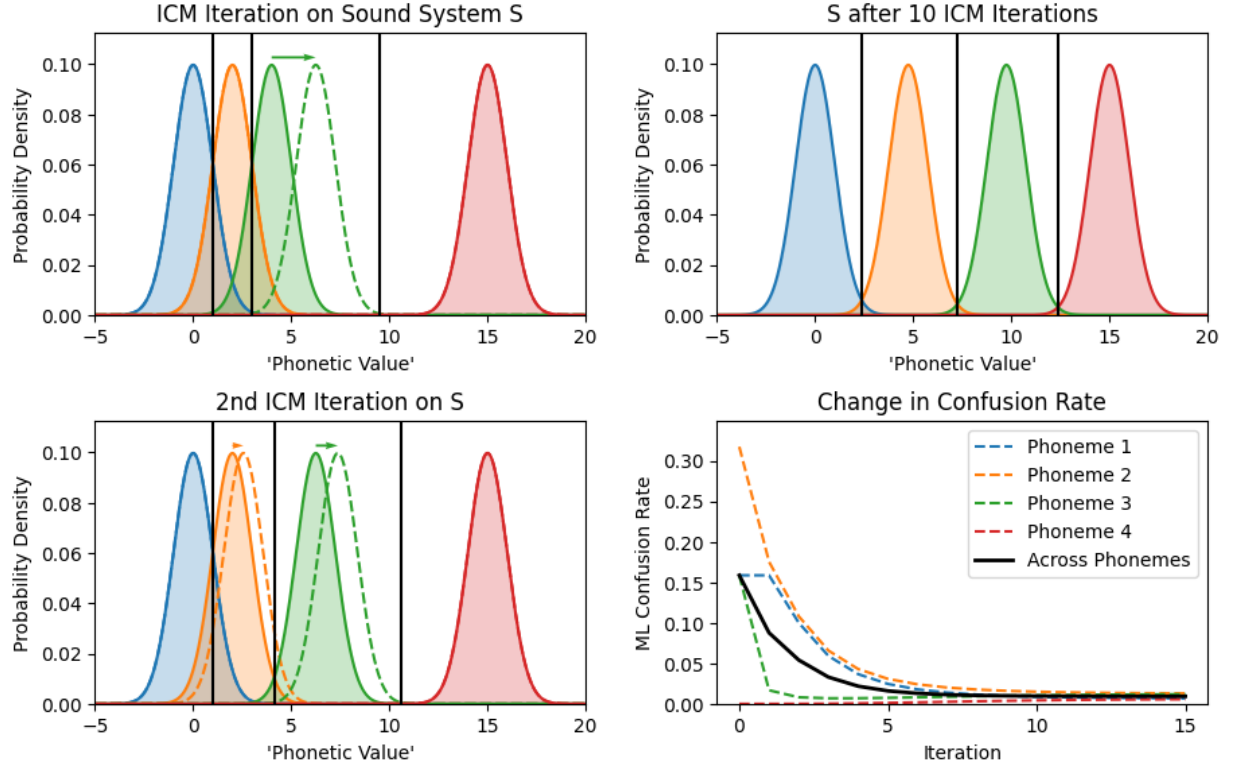


Figure 2.3: An Iterated Confusion Minimization (ICM) simulation on a sound system S with uneven dispersion. **Top Left:** One ICM iteration on S . **Bottom Left:** Updating the ML perceptual boundaries allows a second ICM iteration to take place. **Top Right:** Within 10 iterations, the phonemes of S have incrementally chain-shifted to phonetic values with even spacing. **Bottom Right:** Each iteration lowers the overall ML confusion rate over the system towards an asymptotic minimum around 0.93% for this system.

confusion paradox.

This ICM simulation is generally compatible with the chain shift mechanisms described by Blevins (2004) using exemplar models and Boersma and Hamann (2008) using bidirectional constraint grammars. Since both of these models work fundamentally by restructuring phonetic distributions in a way that reduces confusion, they can be interpreted broadly as framework-specific implementations of ICM. However, abstracting away from specific phonological frameworks to the general mathematical terms of ICM has the benefit of facilitating a more direct analysis of how such mechanisms interact with our quantitative measure of perceptual contrast, i.e. maximum likelihood confusion (see Equation 2.3).

We can see from the bottom right graph of Figure 2.3 that each ICM iteration lowers the overall maximum likelihood confusion rate towards an asymptotic minimum. The property that overall confusion never rises holds in general for any ICM simulation, mathematically guaranteed by the fact that Equations 2.1 and 2.4 encoded into this algorithm define optimal listeners and speakers based on optimization of overall confusion.⁷ This allows for interpretation of ICM as a local optimization algorithm with respect to confusion.

Although ICM always results in lowered overall confusion, it should be noted that ICM is not a complete model of sound change. Other mechanisms of sound change are expected to increase confusion. In fact, the sort of chain shifts ICM can model are commonly triggered by confusion-increasing sound changes such as phonemic mergers. ICM can thus be understood as a restoration mechanism that re-optimizes confusion following upticks in confusion resulting from other sound change mechanisms.

Section 5.1 will show that, like many local optimization algorithms, ICM most often discovers globally optimal solutions, helping to explain the typological dominance of perceptually optimal vowel systems (cf. Liljencrants and Lindblom, 1972; Schwartz et al., 1997b) as an emergent result of a realistic sound change mechanism, despite common characteri-

⁷However, optimal listener behavior does *not* necessarily optimize phoneme-level confusion, e.g. Phoneme 4's confusion increases in Figure 2.3 even as overall confusion decreases.

zations of sound change as lacking in any explicit goal (cf. Blevins, 2004). Also like other local optimization algorithms, however, ICM may sometimes fail to find global optima due to susceptibility to local optima. Section 5.2 will show that these local optima with respect to the vowel space generally correspond to attested yet non-dominant systems and emerge in ICM simulations at rates comparable to observed typological frequency.

CHAPTER 3

Implementing ICM over the Vowel Space

Implementing ICM over the vowel space requires a number of modeling decisions, most having to do with determining the set of possible vowel distributions under consideration (corresponding to Φ in Algorithm 1). Additionally, attempts to use ICM to shed light on vowel system typology requires selection of a vowel system taxonomy that can be used to classify ICM outputs and compare them to a typological baseline. These decisions are explained in the following sections.

3.1 Defining the Vowel Space

Following the precedent of previous work within Dispersion Theory (e.g., Liljencrants and Lindblom, 1972; Schwartz et al., 1997b), the current work is concerned only with steady-state vowels (i.e. monophthongs), modeled as coordinates in a two-dimensional formant space. In particular, we will consider the vowel space plotted in Figure 3.1 with boundaries defined using functions adapted from Liljencrants and Lindblom (1972):

$$F1_{min} \leq F1 \leq F1_{max} \quad (3.1)$$

$$F2 \leq F2_a + (F2_{max} - F2_a) \sqrt{\frac{F1_{max} - F1}{F1_{max} - F1_{min}}} \quad (3.2)$$

$$F2 \geq F2_a + (F2_{min} - F2_a) \sqrt{\frac{F1_{max} - F1}{F1_{max} - F1_{min}}} \quad (3.3)$$

where the parameters $F1_{min} = 314$, $F1_{max} = 699$, $F2_{min} = 889$, $F2_{max} = 2164$, and $F2_a = 1373$ (F2 of the lowest vowel) correspond to the extreme points of the vowel space, all

measured in Hz.¹

For sake of faithfulness to empirical data, these parameter values are taken from the average formant values reported in Becker-Kristal (2010)’s acoustic typology for the corner vowels of the /ieaou/ system, the world’s most common vowel system. Simulations not reported here based on the vowel space used by Liljencrants and Lindblom (1972) and Schwartz et al. (1997b) yielded similar results. Simulations using differently sized vowel spaces for different inventory sizes in accordance with Becker-Kristal (2010)’s report of space size correlated with inventory size yielded better fit to his acoustic typological data, e.g. predicting [ɾɐʊ] rather than [iaʊ], but only a single vowel space size setting is reported here for simplicity since system classifications (see Section 3.3) were largely unaffected.

Although distance-based energy functions used to evaluate perceptual contrast in previous dispersion models are sensitive to the particular acoustic scale used, the overlap-based metric of maximum likelihood confusion does not depend on scale (see Appendix A). Even though past dispersion models have generally used psychoacoustic scales like Mel or Bark, ICM simulations over the vowel space in coming sections use the Hz scale because the empirical studies that inform their phonetic distribution models (see Section 3.2) overwhelmingly report acoustic data in Hz.

Similarly, whereas past dispersion models have generally used $F2'$, corresponding to $F2$ perceptually adjusted for $F3$ and $F4$ (Carlson et al., 1970, 1974), the current work simply uses raw $F2$ values due to the limited availability of acoustic studies reporting standard deviation for higher formants.

Although ICM simulations require a roughly continuous phonetic space, some level of discretization is a practical necessity for computer implementation. In the current work, 101 distinct possible values are used for both $F1$ and $F2$, yielding a granularity of 3.85 Hz in $F1$ and 12.74 Hz in $F2$. This is comparable to or better than the 10 Mel granularity used by

¹Liljencrants and Lindblom (1972) used $F1_{min} = 350$, $F1_{max} = 850$, $F2_{min} = 800$, $F2_{max} = 1700$, and $F2_a = 1150$, all in the Mel scale and with $F2'$ instead of $F2$.

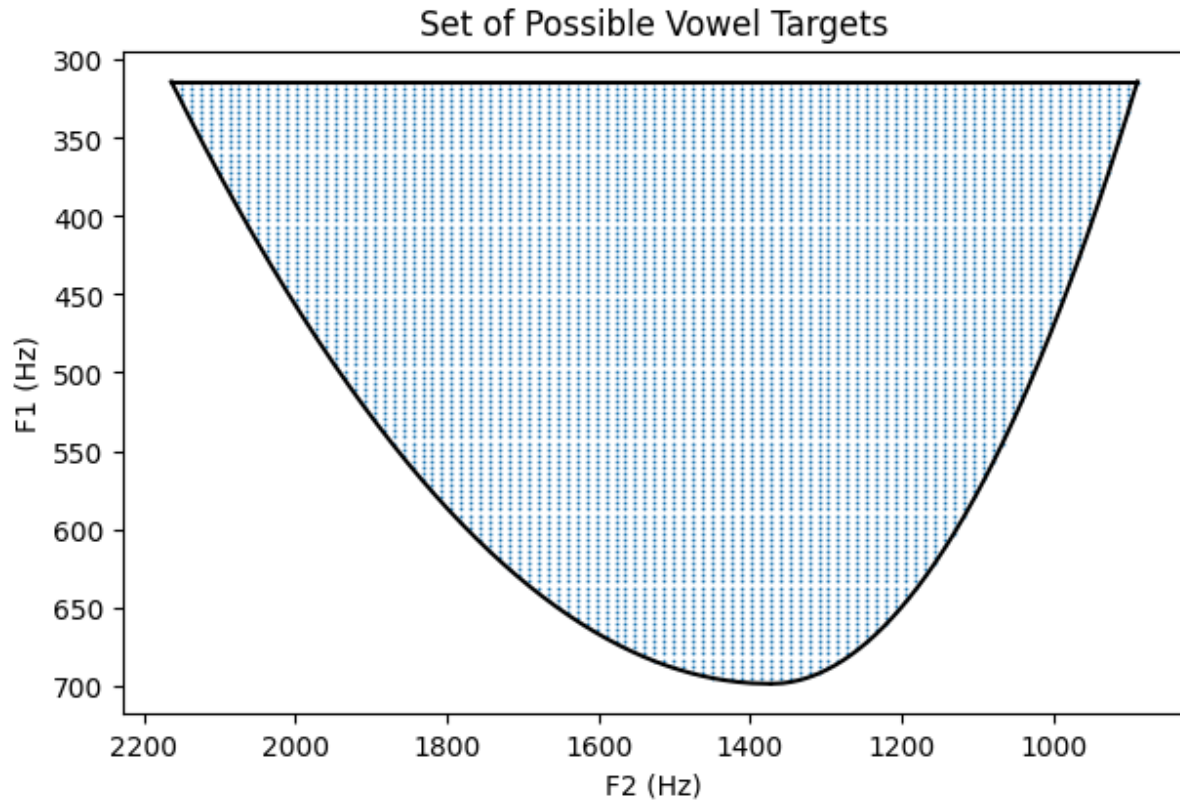


Figure 3.1: The vowel space defined by Equations 3.1-3.3. The 6,719 vowel targets considered by the current ICM simulations are plotted to give a sense of granularity. Any appearance of uneven spacing is the result of plotting imprecision not reflective of the simulations.

Liljencrants and Lindblom (1972) and much more fine-grained than the gaps between the 39 candidate vowels used by Schwartz et al. (1997b), as large as 87 Hz in F1 and 417 Hz in F2. The resulting 6,719 possible vowel targets within the vowel space boundaries set by Equations 3.1-3.3 are plotted in Figure 3.1.

3.2 Constraining the Set of Possible Phonetic Distributions

Whereas the energy functions used in past work are well-defined for a given vowel system given only the prototypical formant coordinates of the vowels making up that system, maximum likelihood confusion rate and hence ICM are only well-defined if the distribution shape associated with a given vowel target is known. For this reason, ICM over the vowel space requires a notion of the possible set of vowels much richer in phonetic detail than that used in past dispersion models.

The simplest approach, the one taken here, treats distribution shape as a one-to-one function of vowel target. In particular, each vowel will be assumed to take an independent bivariate normal distribution determined as a function of a mean target value at some F1-F2 coordinate in the vowel space defined in the last section. The standard deviations in both dimensions are defined as a function of these mean values based on linear regression models fit to by-phoneme formant mean and standard deviation data drawn from acoustic studies of a number of different languages, reported in Equations 3.4-3.7 below. A more thorough evaluation of the level of idealization involved in this approach is given in Section 6.2.

Because level of phonetic variance varies significantly across speech contexts, two statistical models were used: one fit to word list speech data and one fit to running speech data. Since these represent two endpoints along a continuum of speech with low and high phonetic variance respectively², similar results across these two settings would indicate that ICM is broadly robust against the particular standard deviation model used. The standard

²Although even higher phonetic variance occurs in naturalistic non-laboratory speech.

deviation models based on word list and running speech will hereafter be referred to freely as the WL and RS models for sake of brevity.

The data set on which the standard deviation models are based was created by combining through the studies referenced in Becker-Kristal (2010)’s acoustic typology (which did not report standard deviations) and extracting mean and standard deviation data for each phoneme from these sources. Since most acoustic studies tend not to report standard deviation information, a final data set of 33 sources much smaller than Becker-Kristal’s corpus was initially considered. Since higher formants are less commonly reported, only F1 and F2 standard deviations are modeled.

Since studies that pool together multiple speakers without conducting speaker normalization (cf. Adank et al., 2004) risk artificially inflating the variances of individual phonemes, the data set was further narrowed down to only include studies which report mean and standard deviation data separately for each speaker.

The WL model fit to word list speech yielded the following equations:³

$$\sigma_1 = 14.72288 + 0.04928\mu_1 \quad (3.4)$$

$$\sigma_2 = 88.5 - 0.06765\mu_1 - 0.001427\mu_2 + 0.00003869\mu_1\mu_2 \quad (3.5)$$

The RS model fit to running speech with a random effect for speaker included 6 speakers across 4 languages, and yielded the following equations:

$$\sigma_1 = 19.82 + 0.1490\mu_1 + 0.01200\mu_2 - 0.00002057\mu_1\mu_2 \quad (3.6)$$

$$\sigma_2 = 144.9 + 0.01052\mu_1 + 0.02892\mu_2 - 0.00009142\mu_1\mu_2 \quad (3.7)$$

In both of these models, the parameters μ_1 , μ_2 , σ_1 , and σ_2 denote F1 mean, F2 mean, F1 standard deviation, and F2 standard deviation respectively, all measured in Hz.

³The WL model given here was fit before completion of the data set. Simulations with an updated WL model, not yet run due to time constraints, are planned for the near future. Results in Chapter 5 suggest ICM output patterns are robust against the particular model used, so a drastic change in results with an updated model seems unlikely.

Additionally, the simulations of the current work experiment with scaling up F2 standard deviations given by σ_2 in Equations 3.5 and 3.7, the overlap-based analog of scaling down distances, which past dispersion models have found necessary to avoid over-prediction of high non-peripheral vowels (e.g., ten Bosch, 1991; Schwartz et al., 1997b). A constant scaling factor was used for all F2 standard deviations within a given ICM simulation, taking 31 possible values at 0.1 intervals from 1.0 to 4.0.

For sake of brevity, ICM parameter settings are given more concise names in subsequent figures and discussion, e.g. the WL 2.1 model for simulations using the WL model with F2 scaling set to 2.1.

3.3 Classifying Vowel System Configurations

Comparing the outputs of ICM simulations to typological data requires a taxonomy into which ICM outputs can be classified as well as a typological baseline providing the observed frequency with which the systems of this taxonomy occur in natural languages.

The typological baseline used here derives from Schwartz et al. (1997a)’s analysis of the primary vowel systems of Maddieson (1984)’s UPSID (UCLA Phonological Segment Inventory Database), but their taxonomy has been recast into the classification scheme used in Becker-Kristal (2010)’s acoustic typology. Becker-Kristal (2010)’s taxonomy has the advantage that it can easily be implemented in the form of a computer program, a practical necessity for analyzing the outputs of the 62,000 ICM simulations of the current work. By contrast, Schwartz et al. (1997a)’s taxonomy makes a number of fine-grained distinctions between non-peripheral vowels that would require hard-coding of acoustic cut-off points between different phoneme transcriptions. Furthermore, Becker-Kristal (2010) analysis suggests that several of these distinctions rest on shaky acoustic foundations anyhow, e.g. Schwartz et al. (1997a)’s distinction between /ieaoui/ and /ieaouə/ appears to be an artifact of inconsistent transcription choice within UPSID for a common non-peripheral vowel

more properly transcribed [9].

Although Becker-Kristal (2010)’s taxonomy is comparable in granularity to Crothers (1978)’s better known taxonomy, the latter is not suitable for computer implementation over ICM outputs because of its idiosyncratic treatment of ‘defective’ vowel systems, which deviate in arrangement of peripheral vowels relative to dominant systems. Each defective system would need to be individually hard-coded into a computer program, and it is not possible to systematically generalize Crothers (1978)’s taxonomy to defective systems potentially output by ICM not present in his data set.

Given the necessity of taxonomy conversion, Schwartz et al. (1997a)’s taxonomy is the preferred source on the basis of its use as the baseline for the models of Schwartz et al. (1997b) and de Boer (1999) as well as the widespread use and greater reliability of UPSID, on which it is based.

Becker-Kristal (2010) assigns each system a label of the format $\#_p X \#_{np}$, where $\#_p$ denotes the number of peripheral vowels in the system, X is given the value L, R, or S for left-crowded, right-crowded, or symmetric inventories, and $\#_{np}$ denotes the number of non-peripheral vowels. A system is considered left- or right-crowded if it has more peripheral vowels to the front or back of its lowest vowel, e.g. 4L0 /ie-a-u/ vs. 4R0 /i-a-ou/, and symmetric otherwise, e.g. 3S0 /i-a-u/ and 5S1 /ie-a-ou-ə/. Additionally, inventories with two or more non-peripheral vowels are given the suffix ‘a’ if these vowels are arranged vertically, e.g. 5S2a /ie-a-ou-yø/, and ‘b’ if the system includes a horizontal contrast between two high non-peripheral vowels, e.g. 5S3b /ie-a-ou-ywø/.⁴

Typological frequencies based on reclassification of Schwartz et al. (1997a)’s taxonomy into that of Becker-Kristal (2010) are given in Table 3.1. Details of this conversion between taxonomic systems are given in Appendix B.

⁴Becker-Kristal (2010) uses ‘a’ only where necessary for disambiguation with ‘b’-type configurations in his data set, but the current work generalizes ‘a’ for systematicity.

Inventory Size	System	Transcription	Count	Frequency
3 (n=19)	3S0	/iau/	19	1.000
4 (n=25)	4L0	/ieau/	14	0.560
	3S1	/iaua/	7	0.280
	4R0	/iaou/	4	0.160
5 (n=109)	5S0	/ieaou/	97	0.890
	4L1	/ieaui/	5	0.046
	4R1	/iaoui/	5	0.046
	5L0	/ieeau/	1	0.009
	3S2a	/iauiə/	1	0.009
6 (n=60)	5S1	/ieaouə/	41	0.683
	6L0	/ieeaou/	14	0.233
	6R0	/ieaou/	5	0.083
7 (n=44)	7S0	/ieeaou/	23	0.523
	5S2a	/ieaouiə/	13	0.295
	6L1	/ieeaouə/	8	0.182
8 (n=19)	7S1	/ieeaouə/	8	0.421
	5S3b	/ieaouya/	4	0.211
	8L0	/ieeaou/	3	0.158
	8R0	/ieeaou/	2	0.105
	6L2a	/ieeaouya/	1	0.053
	5R3a	/iaouyø/	1	0.053

Inventory Size	System	Transcription	Count	Frequency
9 (n=24)	7S2a	/ieeaouia/	7	0.292
	9S0	/ieeaouu/	7	0.292
	7L2a	/ieeaouia/	4	0.167
	8L1	/ieeaouia/	2	0.083
	6R3b	/ieaouya/	2	0.083
	6R3a	/ieaouya/	1	0.042
	5S4b	/ieaouya/	1	0.042
10 (n=8)	8L2a	/ieeaouia/	2	0.250
	9S1	/ieeaouuə/	2	0.250
	10L0	/ieeaouu/	1	0.125
	9L1	/ieeaouia/	1	0.125
	7S3a	/ieeaouya/	1	0.125
	6L4b	/ieeaouya/	1	0.125
11 (n=4)	8L3a	/ieeaouya/	2.67	0.667
	8R3a	/ieeaouya/	1.33	0.333
12 (n=2)	10L2a	/ieeaouuə/	1	0.500
	7S5b	/ieeaouia/	1	0.500
13 (n=1)	10L3a	/ieeaouuia/	1	1.000
14 (n=1)	8L6a	/ieeaouya/	1	1.000
15 (n=1)	9L6b	/ieeaouya/	1	1.000

Table 3.1: Typological frequency (split by inventory size) for each system based on recasting of Schwartz et al. (1997a)’s analysis of the primary vowel systems of UPSID into Becker-Kristal (2010)’s taxonomy. A possible transcription for each system is given in IPA, although it should be noted that most system labels collapse together a variety of possible transcriptions.

CHAPTER 4

Example ICM Simulations over the Vowel Space

Finally, we have all the pieces in place to run ICM simulations over the vowel space and compare the results to a typological baseline. The next two sections present example simulations illustrating the behavior of ICM using simulations over 6 vowels. Section 4.1 shows that ICM is able to predict chain shifts along the vowel space periphery, a commonly observed pattern of sound change. Section 4.2 shows that ICM is able to predict non-peripheral vowels interior to the vowel space, i.e. not on the high periphery, a systematic shortcoming of past dispersion models. Both simulations use the WL 2.0 model of possible vowel distributions (see Section 3.2).

4.1 Predicting Chain Shifts

Figure 4.1 shows an ICM simulation over a randomly initialized 6-vowel system. The top two graphs in the figure show the initial and final state of the system. In each of these, the white quasi-triangular shape demarcates the bounds on the space of vowel targets considered in the simulation (see Equations 3.1-3.3), the blue circles indicate these targets, and the magenta lines indicate the maximum likelihood perceptual boundaries between each category.

These graph features are overlaid on a heatmap showing the probability density of vowel tokens at each value, with lighter shade corresponding to higher probability density. This visualization of probability density does not differentiate token frequency by phoneme (the color dimension previously used for this purpose in one-dimensional simulations is now used

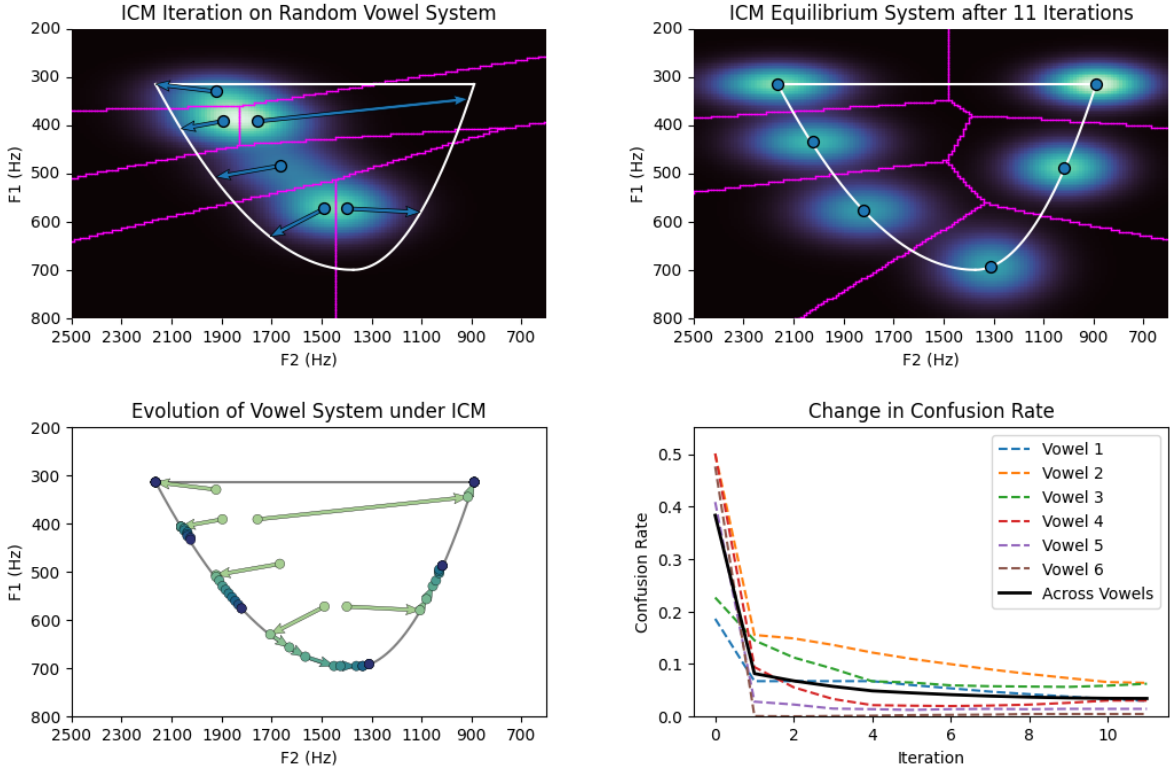


Figure 4.1: An Iterated Confusion Minimization (ICM) simulation on a randomly initialized 6-vowel system. **Top Left:** One ICM iteration. **Bottom Left:** The evolution of the system under ICM demonstrates a chain shift along the vowel space periphery. **Top Right:** At 11 iterations, the vowels have reached equilibrium at the second most typologically frequent system 6L0 /ieeaou/. **Bottom Right:** Each iteration lowers the overall ML confusion rate towards an asymptotic minimum around 3.44% for this system.

for probability density), such that the heavily overlapping initial distributions of the top left cannot be visually distinguished. However, each distinct distribution can clearly be resolved in the top right, which shows a highly-dispersed system with low overlap that results from ICM over the initial vowel system of the top left.

As in the earlier one-dimensional simulations, the ICM output of the top right is stable because it is self-optimal, which we can confirm visually from the fact that each vowel target appears as far away as possible from its perceptual boundaries. By contrast, the initial system is not self-optimal, and the arrow leading out from each initial vowel target shows the change in phonetic target that will minimize confusion against existing perceptual boundaries in magenta.

The bottom left plot shows the evolution of the system under ICM, with each dot representing a vowel target at some iteration, each arrow representing the optimization of that target, and color of each dot and arrow representing the associated iteration number, with darker shades corresponding to later iterations. After some relatively large shifts in the first iteration due to the extremely poor dispersion of the initial system, subsequent iterations tend to involve much smaller steps. The bottom right figure shows that the overall confusion rate only ever goes down, approaching an asymptotic minimum as we observed in the one-dimensional simulations.

This simulation is notable for its illustration of a chain shift along the vowel space periphery, which is clearly observed in the bottom left graph of Figure 4.1. This pattern, which is common among ICM simulations, has particular appeal given the ubiquity of chain shifts along the vowel space periphery in the historical phonology of the world’s languages (cf. Labov, 1994, p. 113-291). This chain shift of this simulation in particular appears reminiscent of the Low-Back-Merger Shift of several North American English dialects such as Canadian and California English, in which raising of the low back vowel /ɑ/ as part of its merger with /ɔ/ is followed by lowering and retraction of front vowels (Labov et al., 2006).

Although the 6L0 system /ieeaou/ output by this simulation is a relatively common

attested 6-vowel system, this is not the typologically dominant system, i.e. 5S1 /ieaouə/. Furthermore, we will see in Section 5.1 that this 6L0 system is neither the most frequent nor the optimal ICM output under most parameter settings. The reason for the stability of this system is because it is a *local optimum*, a common pitfall of local optimization algorithms like ICM.

The visualization of the equilibrium 6L0 system /ieəaou/ output by this simulation gives an intuition for how a peripheral system like this can be stable even though it is not a global optimum. The perceptual boundaries associated with each vowel in this peripheral system form ‘cake slice’-shaped perceptual regions pointed towards the interior of the vowel space. In order to get to the globally optimal 5S1 system, one of these vowels would have to move towards one of its perceptual boundaries in the vowel space interior. However, this would result in a rise in confusion given the preexisting perceptual boundaries, so the ideal speaker/listener, whose behavior is locally optimizing rather than directed towards the goal of a true global optimum, can easily get stuck in this local optimum.

4.2 Predicting Interior Vowels

Figure 4.2 shows an ICM simulation with a different initial system and final output from those of Figure 4.1. The behavior of the simulation is broadly similar to that of the previous simulation, with a large initial shift of vowels away from a highly unnatural initial system with low dispersion, followed by incremental shifts towards the final system associated with a reduction in overall confusion rate towards an asymptotic minimum. This minimum in confusion turns out to be a global optimum better than the local optimal output of Figure 4.2 (maximum likelihood confusion 0.027 vs. 0.034).

The final system configuration shown in the top right of the ICM simulation of Figure 4.2 is an instance of the 5S1 /ieaouə/ system, far and away the dominant 6-vowel system accounting for 68.3% of primary 6-vowel systems in UPSID (cf. Table 3.1). This result

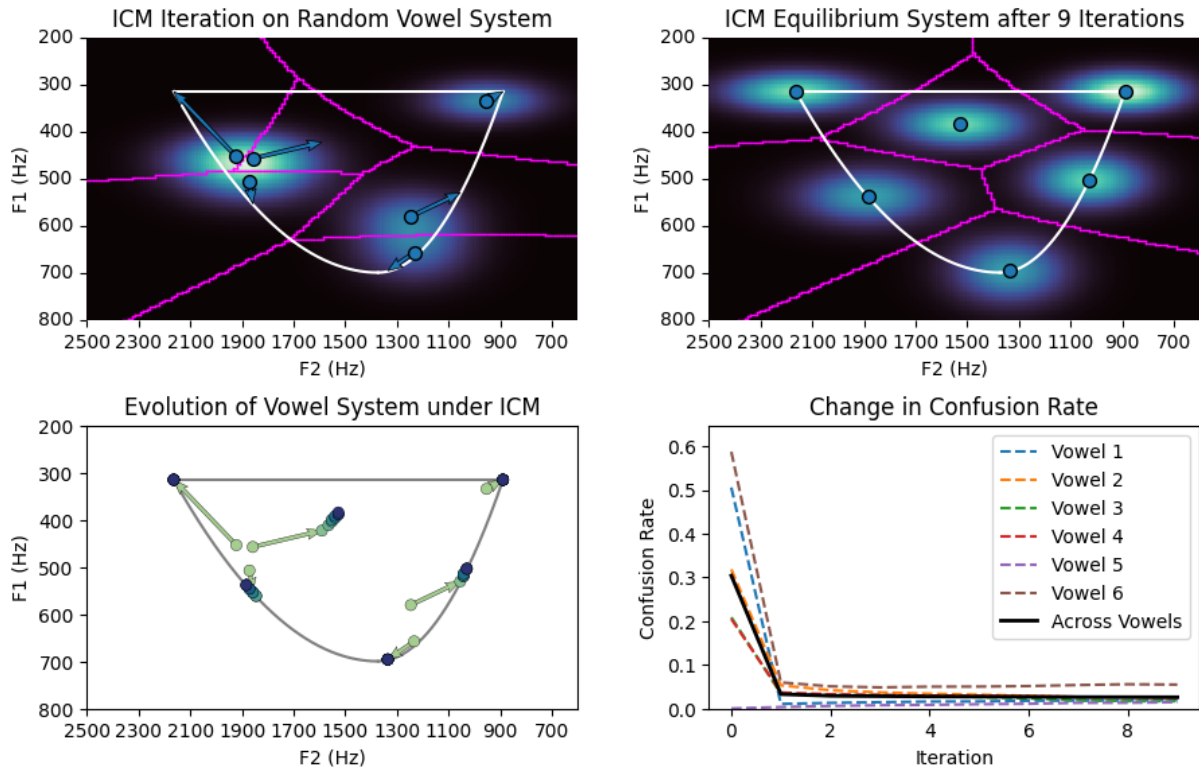


Figure 4.2: An ICM simulation over a randomly initialized system 6-vowel system different from that of Figure 4.1 results the typologically dominant 5S1 system /ieaouə/.

is attractive not only because it demonstrates how ICM can transform even a grotesquely unnatural system with low dispersion into a typologically dominant vowel system with high dispersion, but also because of the narrower acoustic structure of this output system. In particular, the central vowel has a formant value of roughly [ə] intermediate in height between the mid and high vowels [e̞ ɤ] and [i u], consistent with Becker-Kristal (2010)’s acoustic typology and in contrast to dispersion models that only predict a fully high [i] (e.g., Liljencrants and Lindblom, 1972; Schwartz et al., 1997b).

Beyond greater acoustic accuracy for this particular 5S1 system, the prediction of [ə] highlights a further strength of ICM over more conventional dispersion models using distance-based energy functions. Namely, ICM can predict interior vowels like /ə/ not well explained in those models. Liljencrants and Lindblom (1972)’s original simulations predicted no interior vowel for inventories smaller than 10, and no parameter setting of Schwartz et al. (1997b)’s model reported for inventory sizes up to 7 yields an interior vowel. This systematic failure results from the fact that interior vowels like /ə/ have a relatively small acoustic distance to *all* other vowels, resulting in a higher energy function for systems with interior vowels relative to systems without.

By contrast, the top right visualization of Figure 4.2 gives an intuition for how interior vowels can be stable under ICM. The interior vowel of this 5S1 system is in effect fenced into the vowel space interior by its perceptual boundaries in each direction with /ieou/. Because the optimal target for a given set of perceptual boundaries is generally maximally far away from those boundaries, a system including an interior vowel target surrounded by vowels on all sides can be stable if that interior vowel target is centered between its perceptual boundaries with those surrounding vowels, as is the case here.

Indeed, the results of Section 5.3.4 will show that a 5S1 configuration with an interior central vowel is not only stable, but also globally optimal when using maximum likelihood confusion (see Equation 2.3) as an evaluation metric. Becker-Kristal (2010, p. 152) showed that this configuration is predicted by maximizing the smallest pairwise distance rather than

minimizing an energy function that takes into account all pairwise distances, an approach also taken by ten Bosch (1991) and Flemming (2005).

Although this approach may seem ad hoc within the tradition of energy-based dispersion models, the overlap-based nature of ICM gives a more intuitive, principled basis for this sort of local approach to dispersion. In particular, the visualizations of Figure 4.2 make clear that an optimal listener/speaker should care far more about the distance between acoustically adjacent vowel pairs, since these have far more phonetic overlap than non-adjacent pairs. Why, for instance, should the distance between /a/ and /ə/ play any role in evaluating the level of perceptual contrast in this system (as assumed by conventional energy functions) when these two phonemes have virtually no overlap and do not even share a perceptual boundary?

CHAPTER 5

Generating Vowel System Typology

In order to test the ability of ICM over the vowel space to generate vowel system typology, a large number of simulations were run over randomly initialized vowel systems. These simulations differed based on three parameters: inventory size, standard deviation model, and F2 s.d. scaling factor.

The simulations cover 10 inventory sizes from 3 to 12, which is the same set of inventory sizes simulated by Liljencrants and Lindblom (1972) and corresponds to every inventory size represented by the primary vowel systems of at least two languages in UPSID as per Schwartz et al. (1997a). One of the two standard deviation models given in Section 3.2, the WL and RS models based on word list and running speech respectively, was selected in each simulation. Finally, 31 F2 s.d. scaling factors at 0.1 intervals from 1.0 to 4.0 were considered (cf. Section 3.2). This resulted in 620 ($= 10 \times 2 \times 31$) parameter settings. 100 simulations were conducted for each parameter setting, for a grand total of 62,000 simulations.

The next Section 5.1 summarizes the most frequent ICM outputs across parameter settings, showing that ICM most often results in perceptually optimal systems and outperforms the past baseline set by Schwartz et al. (1997b) in predicting typologically dominant systems as per Table 3.1 for larger inventories.

Given significant variation in both typology and ICM outputs, Section 5.2 evaluates the correlation of ICM emergence rate with typological frequency and compares this to the baseline set by de Boer (1999, Chapter 6).

A more detailed presentation of simulation results split by inventory size is given in

5.1 Predicting Typologically Dominant Systems

In attempting to account for the typologically most frequent vowel systems, models in Dispersion Theory such as that of Schwartz et al. (1997b) are considered to predict a vowel system if that system has an optimal energy function under a given parameter setting. ICM more directly predicts a system will be most frequent in typology if it is most frequent among simulation outputs. However, since ICM can be understood as a local optimization algorithm with respect to maximum likelihood confusion (see Equation 2.3), it is natural to ask what vowel system is optimal with respect to this metric for a given parameter setting and to compare this optimum against ICM outputs (which are not guaranteed to be global optima) and typology.

Figure 5.1 identifies the most frequent ICM output and Figure 5.2 identifies the perceptually optimal vowel system across parameters settings.¹ The rows of each figure are analogous to the so-called ‘phase space’ diagrams reported by Schwartz et al. (1997b), although the phase spaces across inventory sizes given here can be condensed into a single figure by representing inventory size on the y -axis (which Schwartz et al. (1997b) require for representing the effect of their focalization weight α). The systems in the figures are color coded green if typologically dominant, yellow if attested, and red if unattested in UPSID based on Table 3.1. It should be noted that some systems referred to as unattested here have been reported in sources outside UPSID, e.g. Becker-Kristal (2010) reports instances of 6L3b, 7S3b, and 9S3a, among other systems shown in red in Figures 5.1 and 5.2.

¹Due to the size of the search space (e.g. over 100 quadrillion possible 5 vowel systems over the 6,719 possible vowel targets plotted in Figure 3.1), it is not computationally tractable to find the global optimum through an exhaustive search of this space, so Figure 5.2 reports the optimal vowel system among the 100 ICM outputs at each parameter setting. We can nevertheless be fairly confident that each reported optimum corresponds to the true global optimum, since the probability that a local optimization algorithm finds the global optimum approaches 1 with a large number of randomly initialized runs.

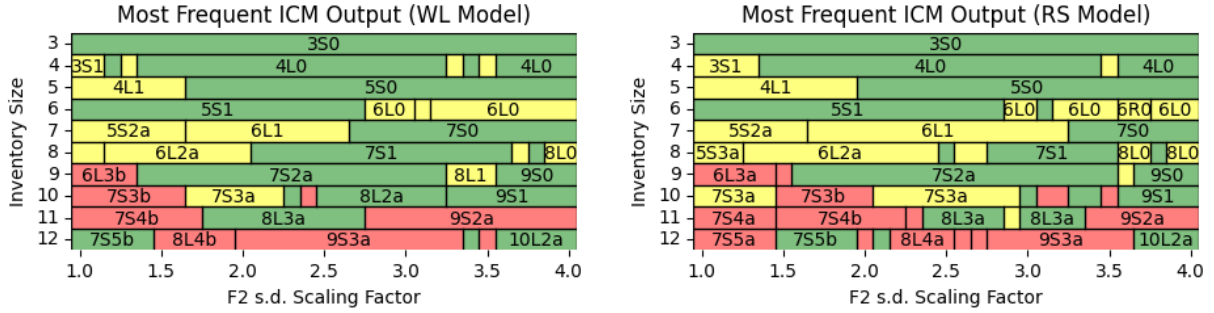


Figure 5.1: Most frequent vowel system output by ICM for each parameter setting, shaded green if dominant, yellow if attested, and red if unattested in UPSID (see Table 3.1).

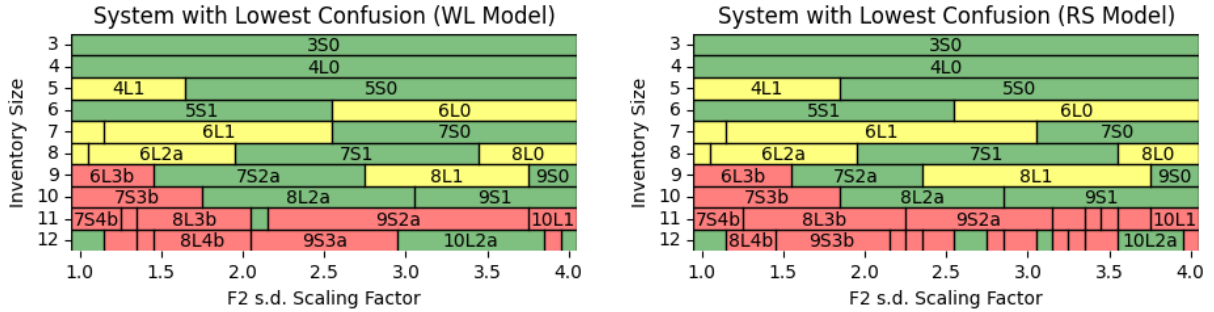


Figure 5.2: Perceptually optimal vowel system with lowest maximum likelihood confusion under each parameter setting, shaded by typological attestation following Table 3.1.

The broad agreement between the most frequent ICM outputs of Figure 5.1 with the optimal systems of Figure 5.2 is immediately striking. Although local optimization algorithms are not in general guaranteed to find the global optimum most of the time (and indeed there are minor regions of disagreement between the figures where ICM does not), ICM does so under the clear majority of parameter settings. In contrast to past dispersion models largely divorced from mechanism, the fact that a plausible model of sound change like ICM produces perceptually optimal systems most of the time gives a clear mechanical explanation rooted in diachrony for the success of Dispersion Theory in accounting for typologically dominant vowel system structures.

Despite some differences between the WL and RS model results, we can observe that the broad pattern of dominant ICM outputs and optimal vowel systems is robust against the particular standard deviation models used. Across standard deviation models and inventory sizes, systems with stronger F2 scaling tend to have more peripheral vowels (corresponding to the first number of each system label), similar to past results from Schwartz et al. (1997b). ‘Phase transitions’ between systems tend to require stronger F2 scaling for the RS model than the WL model, e.g. 6L1 transitions to 7S0 at 2.7 for the WL model but 3.4 for the RS model in Figure 5.1, although the reason for this is not obvious.

For up to 8 vowels, no unattested system has the highest ICM emergence rate under any parameter setting. This contrasts with Schwartz et al. (1997b)’s model, which for 6 and 7 vowels predicts unattested 4L2b [iæɒuyʊ] and 5S2b [iɐɒɒuyʊ] systems given weak F2 scaling ($\lambda > 0.7$ and $\lambda > 0.4$ respectively) and which is not reported for 8 or more vowels.²

Although Schwartz et al. (1997b) are able to predict every dominant vowel system for inventory sizes up to 7, no parameterization of their model is able to do so simultaneously. In particular, the dominant 7-vowel system 7S0 /ieɛəɒu/ requires F2 scaling too strong to predict 5S1 /ieaoui/. While the observation that 5S1 and 7S0 are favored by weaker and

²Consulting Becker-Kristal (2010), all dominant systems predicted by ICM are attested up to 10 vowels, as are these 4L2b and 5S2b systems predicted by Schwartz et al. (1997b).

stronger F2 scaling also holds here, Figure 5.1 shows overlap of correct prediction at a single F2 scaling factor value of 2.7 in the WL model.

Indeed, this parameter setting (WL 2.7 model) predicts a dominant vowel system for *every single* inventory size up to 11, although one should note that ‘dominance’ is less meaningful for inventory sizes starting at 10 for which no system occurs more than thrice in UPSID. The remaining dominant systems 9S0 and 9S1, which are tied in typological frequency with the systems 7S1 and 7S2 predicted under this parameter settings, as well as both attested 12-vowel systems are predicted as dominant by ICM using other F2 scalings, with the result that every single typologically dominant vowel system is predicted as dominant by ICM under some setting for both the WL and RS models.

This improvement over past models in predicting larger vowel systems appears owing to the fact that ICM can predict interior vowels, which exist in all systems with labels containing the ‘a’ suffix including the dominant 9, 10, and 11 vowel systems 7S2a, 8L2a, and 8L3a.

5.2 Predicting Typological Frequencies

Since ICM simulations predict variation within inventory sizes, it is possible to evaluate ICM emergence rate of different systems against typological frequency across different parameter settings. Since this involves comparing two probability distributions over the set of possible vowel systems, a useful evaluation metric is a statistical distance measure called Kullback-Leibler (KL) divergence (Kullback and Leibler, 1951), also known as relative entropy.

The KL divergence $D_{KL}(P||Q)$ of a reference distribution P (typological frequencies) from a model distribution Q (ICM emergence rates) is as follows:

$$D_{KL}(P||Q) = \sum_{x \in X} P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad (5.1)$$

where X in this case is the set of possible vowel systems. More intuitively, KL divergence

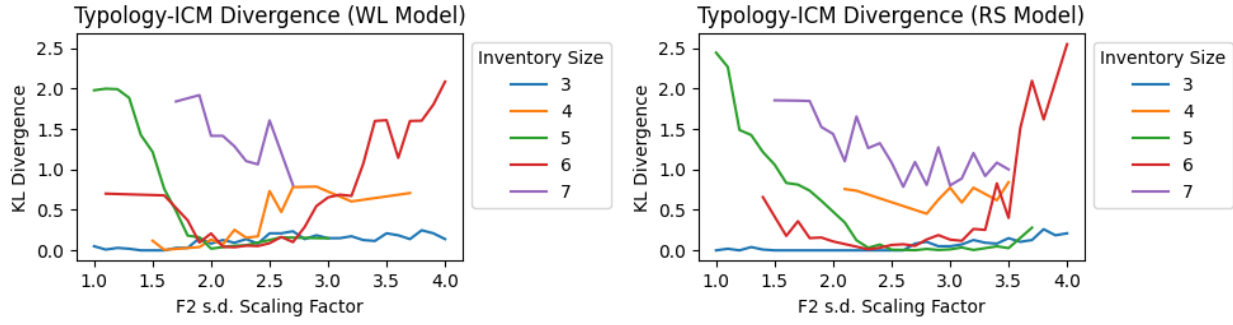


Figure 5.3: KL divergence of typological frequencies given in Table 3.1 from ICM emergence rates across parameter settings.

of typology from ICM emergence rate gives a measure of how ‘surprising’ the observed typological frequencies are when ICM emergence rates are used as a model. A lower value indicates a better model.

Figure 5.3 plots the KL divergence of typology from ICM emergence rate for different inventory sizes across parameter settings. KL divergence is infinite when a model gives zero probability to a value with non-zero reference probability. Because of this, KL divergence is infinity under all parameter settings for inventory sizes where ICM is unable to predict all attested systems simultaneously (even though most attested systems are stable under at least some parameter setting of ICM). This includes all inventory sizes from 8 to 10. It also initially included the inventory size of 5 vowels due to the difficulty of predicting the systems 5L0 /ieɛau/ and 3S2a /iauiə/ simultaneously,³ but since these each exist in less than 1% of 5-vowel systems in Table 3.1 (one system each), they were excluded from the analysis.

The fact that Figure 5.3 includes a finite KL divergence for all inventory sizes 3-7 under certain parameter settings indicates that every single vowel system attested in at least two languages in UPSID with size up to 7 is stable under ICM for these settings.⁴ This covers

³5L0 and 3S2a are also unpredicted by de Boer (1999)

⁴All attested 11- and 12-vowel systems are also stable, but these are excluded from analysis due to scarcity of attestation (four and two systems respectively in Table 3.1).

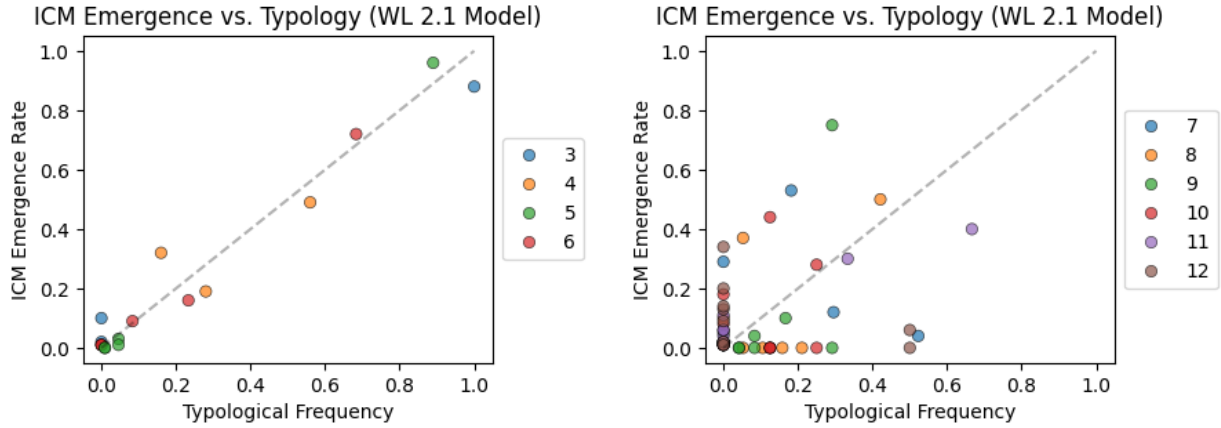


Figure 5.4: ICM emergence rate vs. typological frequency for the WL 2.1 model, split between smaller and larger inventories. Dashed lines correspond to a perfect match.

81% of the primary vowel systems of UPSID.

On first pass, the best divergence across these inventory sizes appears for F2 scaling of 1.9 to 2.3 for the WL model and around 2.3 to 3.1 for the RS model. Inventory sizes 3 through 6 all have good divergence in this range for the WL model, while 4 has significantly worse divergence in the RS model and 7 has markedly worse divergence in both models.

Since it has the lowest maximal KL divergence among inventory sizes 3 to 6 (0.128 for 3 vowels), the WL 2.1 model is taken as the best overall model. Although this setting does not predict the 7S0 /ieɛaou/ as typologically dominant (see Section 5.1), it gives a typologically dominant vowel system the lowest maximum likelihood confusion for every other inventory size up to 11 (see Figure 5.2) and the highest ICM emergence rate for every other inventory size up to 11 other than 10, where it predicts the attested 7S3a system (which loses to ‘dominant’ 8L2a and 9S1 by just one language in Table 3.1). The plots of ICM emergence rate vs. typological frequency for the WL 2.1 model in Figure 5.4 show high correlation for these smaller inventory sizes up to 6 vowels and low correlation for larger inventories, which include a number of predicted unattested systems and unpredicted attested systems, clustered at value 0 on the y - and x -axis respectively.

Although inventories with 6 or fewer vowels account for the large majority of languages (67% of systems in Table 3.1), this correlation breakdown for larger inventories is nevertheless troubling. One possible explanation is that the idealization of a two-dimensional vowel space becomes problematic at 7 vowels. It does not seem coincidental that this is the inventory size at which phonologists begin to use distinctive features beyond height and frontness, e.g. $[\pm\text{ATR}]$ or $[\pm\text{tense}]$ to distinguish pairs like $/e\ \varepsilon/$ and $/o\ \text{ɔ}/$ present in the under-predicted 7S0 system. The possibility of incorporating the phonetic correlates of such features to improve model prediction is left to future work.

Another possibility has to do with simulation of sound change from random initial vowel systems. Given that vowel systems in reality evolve from other natural language vowel systems, which are manifestly *not* random (as evidenced by the very existence of typological tendencies), one should expect a bias in vowel system typology towards systems easily accessed by sound change from other typologically common systems. For example, 7S0 may be systematically under-predicted because the model excludes mechanisms whereby 7S0 evolves from the most common vowel system 5S0 $/ieaou/$, e.g. the canonical 7S0 system, that of Italian, resulted from the collapse of a length contrast in the original dual-length 5S0 system of Latin (Grandgent, 1927, p. 18-19). Capturing this sort of diachronic bias would require a much more complete model of sound change covering phenomena like phonemic merger and split.

Returning to quantitative analysis of emergence rates, KL divergence enables rigorous comparison not only between different ICM models, but also between ICM models and the preexisting baseline set by de Boer (1999, Chapter 6). Table 5.1 shows the KL divergence of typology across inventory sizes from de Boer (1999)’s model, the best ICM model by inventory size, and the best overall ICM model, i.e. the WL 2.1 model.

De Boer (1999) reports frequency of simulation outputs categorized into systems corresponding to Schwartz et al. (1997a)’s taxonomy, so these were reclassified into Becker-Kristal (2010)’s taxonomy following the same procedure used to convert Schwartz et al. (1997a)’s

Inventory Size	KL Divergence		
	de Boer (1999)	Best ICM Model	WL 2.1 Model
3	0.248	0.000 (numerous)	0.128
4	∞	0.007 (WL 1.6)	0.072
5	0.009	0.002 (RS 2.7)	0.041
6	0.028	0.013 (RS 2.3)	0.046
7	0.383	0.786 (RS 2.6)	1.415

Table 5.1: KL Divergence of typology from de Boer (1999)’s results, the best ICM model by inventory size, and the best ICM model across inventory sizes (the WL 2.1 model).

analysis of UPSID into the typological baseline given in Table 3.1 (cf. Section 3.3, Appendix B).

The phoneme labels and corresponding system type labels given by de Boer (1999) were taken at face value in this conversion, but many of his outputs would receive a different label resulting in a worse match with typology if classified directly based on acoustic values. For example, many of his simulation outputs labeled /ieaəou/ have /o/ with a central value approximately [ə] fronter than /ɔ/, which would be classified based on acoustics as already over-predicted 6L1 rather than typologically dominant 7S0. In this way, Table 5.1 evaluates de Boer (1999)’s results more leniently than ICM results.

Since inventory size itself is variable within a given simulation in de Boer (1999)’s model and different inventory sizes are generated through different settings of a noise parameter, there is no exact equivalent of the notion of best overall model across inventory size, which is by contrast a fixed parameter in a given ICM simulation. Since results were reported by de Boer (1999) with whichever noise parameter gave the best results for a given inventory size, a fair comparison likewise takes the best ICM model by inventory size.

When this comparison is made, the best ICM model yields a better KL divergence than

de Boer (1999)’s model for every vowel system size other than 7. Even if using the best overall model with weaker performance for individual inventory sizes, i.e. the WL 2.1 model, ICM still yields a better KL divergence for 3 and 4 vowels, the latter of which has infinite KL divergence in de Boer (1999)’s model due to non-emergence of 4R0 /iaou/. De Boer (1999) also reported simulation results for 8 and 9 vowels, but these likewise had infinite KL divergence due to non-emergence of certain attested systems, also the case for ICM results.

5.3 Detailed Results by Inventory Size.

The following subsections investigate the ICM simulation results for each inventory size in greater depth.

Each subsection includes plots showing the emergence rate of different system configurations for different F2 scaling factors using both the WL and RS standard deviation models. ICM emergence rates are plotted as solid lines and baseline typological frequencies from Table 3.1 are plotted as dashes. Where multiple systems had the same typological frequency, one of these systems is offset at a slightly higher value for visual clarity.

Where relevant, the role of F2 scaling in ICM simulations is compared to the role of F2 scaling in the past model of Schwartz et al. (1997b), which scales F2 using a weight $0 \leq \lambda \leq 1$ on F2 distances in calculation of energy.

Major vowel system configurations up to size 7 each have two additional plots: 1) a plot showing minor variation of vowel targets within the system as circles centered on each vowel target with area equal to emergence rate of that target, and 2) a plot of a representative example of the system showing vowel targets, token probability density, and maximum likelihood boundaries.

These two types of vowel space plots demonstrate results using an F2 scaling factor of 2.2, which was the parameter setting found to have the best correlation between vowel system configuration emergence rate and typological frequency for smaller inventories.

5.3.1 3 Vowels

Figure 5.5 shows ICM results for three vowels. The 3S0 system /iau/ (with possible phonetic variation, e.g. [ɪəʊ]) accounts for 100% of primary 3-vowel systems in UPSID and the overwhelming majority of ICM simulations under all parameter settings. There exist settings of F2 scaling for both the WL and RS models that yield a 100% emergence rate exactly matching the typological baseline. The 3S0 system is also the globally optimal 3-vowel system with respect to maximum likelihood confusion rate.

Within the 3S0 systems produced, there was some variation regarding the exact phonetic output produced. In particular, there were 4 values for /i/, 3 for /u/, and 2 for /a/. This variation is all fairly minor, and the slightly lowered /i/ and /u/ targets are reminiscent of ‘hole’ systems reported in Schwartz et al. (1997a), a distinction not made in the currently used classification scheme.

Figure 5.5 also shows a 3L0 system /iea/, the most frequently emerging non-3S0 system. The plot showing maximum likelihood boundaries gives an intuition for why this system is stable, although the confusion rate is over 30 times worse. Given how few simulations wound up in this configuration, it seems likely that this is not a particularly stable equilibrium, i.e. that even relatively small perturbations to this system would cause it to evolve into a 3S0 system. However, no experimentation with adding perturbations or noise to ICM simulations was conducted for the current work.

5.3.2 4 Vowels

There are three typologically attested 4-vowel systems: 4L0 /ieau/ (56%), 3S1 /iauə/ (28%), and 4R0 /iaou/ (16%). All of these are also produced by ICM simulations, as shown in Figure 5.6.

The typologically dominant 4L0 system has the lowest confusion and highest ICM emergence rate under nearly all parameter settings, with relatively consistent emergence near the

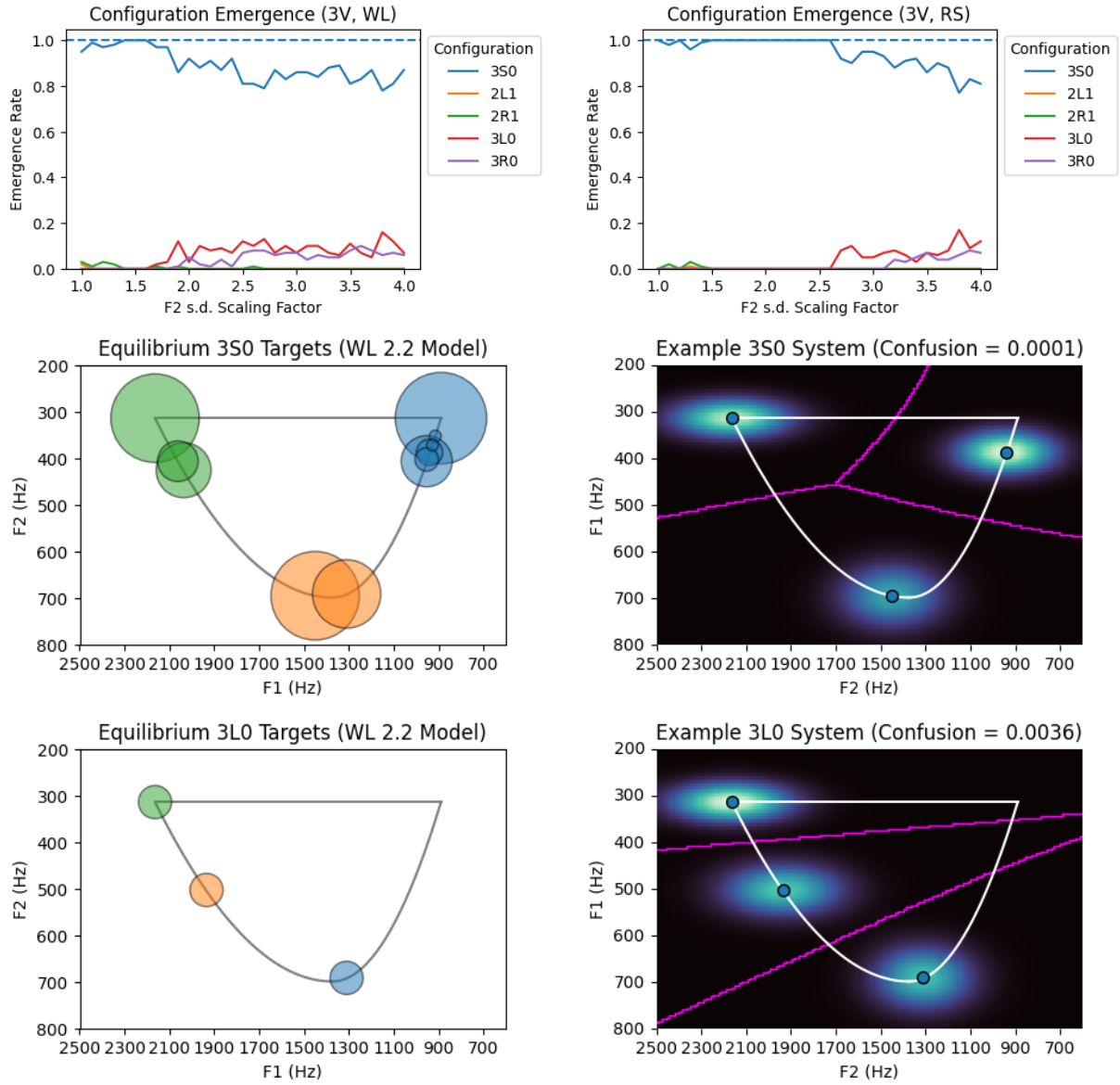


Figure 5.5: ICM results for inventory size 3.

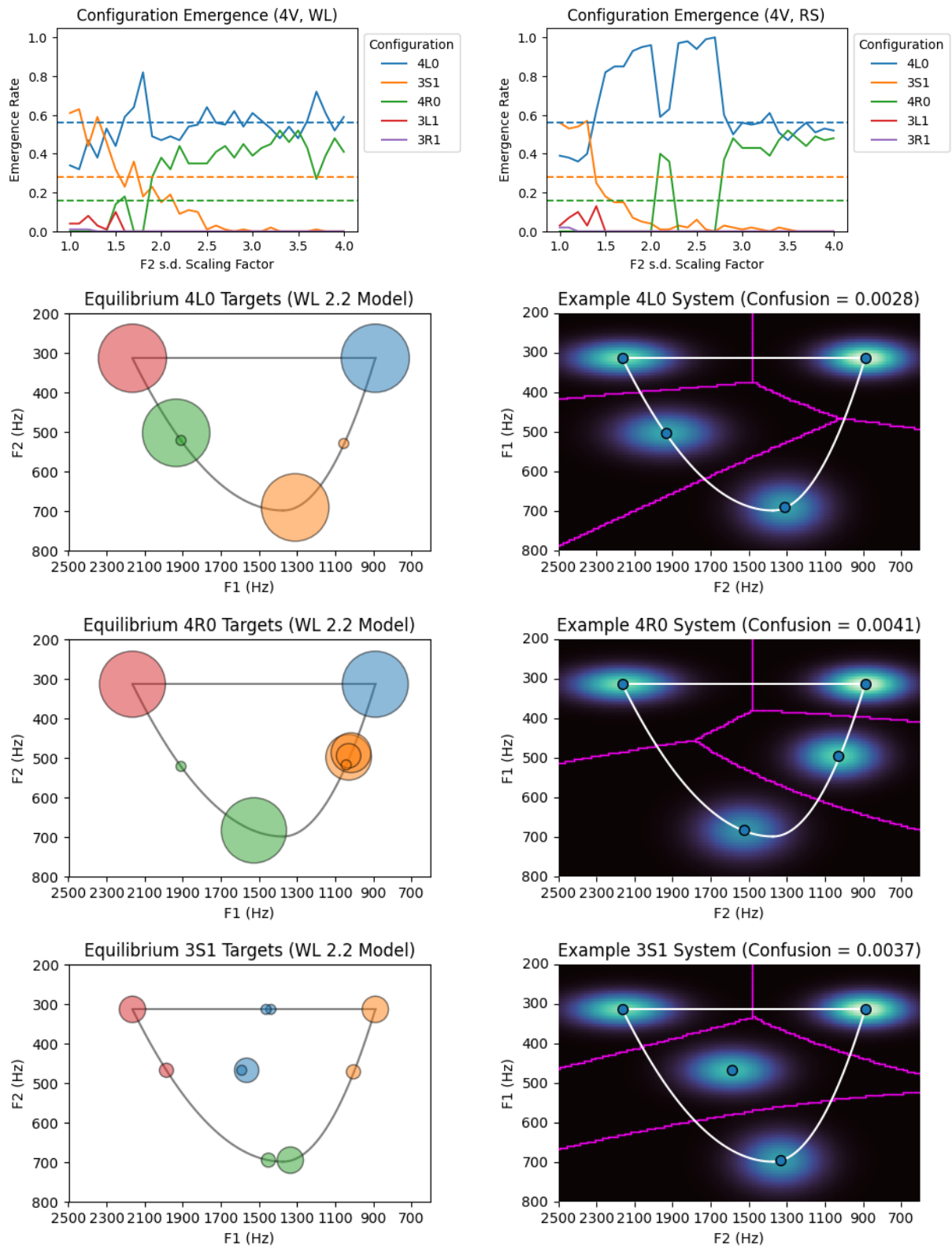


Figure 5.6: ICM results for inventory size 4.

typological baseline (apart from some regions of near-exclusive emergence of 4L0 in the RS model). 3S1 has a high emergence rate with low F2 s.d. scaling factors and in fact out-emerges 4L0 at the lowest scaling factors. 4R0 has a high emergence rate for mid-to-high F2 s.d. scaling factors and only sporadically out-emerges 4L0. This parallels the effect of F2 scaling found by Schwartz et al. (1997b), whose model predicted 3S1 systems with weak scaling ($\lambda > 0.5$) and 4L0 systems with stronger scaling. The best across-the-board agreement between ICM emergence and typological frequency comes for F2 s.d. scaling factors just below 2.0.

With an F2 s.d. scaling factor of 2.2 (displayed in the vowel plots), 3S1 has a lower ICM emergence rate than 4R0 even though 3S1 has a higher typological frequency. Interestingly, however, the optimal 3S1 system has a lower confusion rate than the optimal 4R0 system, so ranking in terms of confusion rate appears to track typological frequency even under parameter settings where the ICM emergence rate ranking does not.

Both the 4L0 and 4R0 systems, normally [ieau] and [iaou], have a single instance of a symmetric subtype approximately [ieou]. Although the majority of ICM 3S1 outputs are [iaua], which matches typological descriptions, an appreciable minority instead have [eaoi], which is not typologically attested.

With a very low F2 s.d. scaling factor, the unattested systems 3L1 /ieɔu/ and 3R1 /eoui/ also occasionally emerged (at low rates), although only attested systems emerged for scaling factors above 1.6.

5.3.3 5 Vowels

5-vowel systems are typologically dominated by the 5S0 system /ieaou/ (89.0%), followed by two minor systems 4L1 /ieaoi/ and 4R1 /iaoui/ (4.6% each) and two rare systems 5L0 /ieəu/ and 3S2a /iauyə/ (0.9%) each attested in only a single UPSID language. Figure 5.7 shows that the dominance of 5S0 also holds in terms of ICM emergence rate under most

settings.

Although the minor systems 4L1 and 4R1 dominate ICM simulations with a low F2 s.d. scaling factor, 5S0 dominates for scaling factors greater than 1.6 for wordlist variances and 2.0 for running speech variances. As was the case for 4-vowel systems, this effect also parallels findings of Schwartz et al. (1997b), who predicted 4L1 systems for weak F2 scaling ($\lambda > 0.5$) and 5S0 systems for stronger scaling.

Apart from parameter settings with insufficient F2 scaling, it is striking just how little variation exists in ICM outputs for 5 vowel systems. Just as 3S0 systems overwhelmingly dominate for 3-vowel systems in both typology and simulation, so too does 5S0 dominate among five vowel systems. This lack of variation is not observed for other inventory sizes in either typology or the ICM simulations of the current work. The agreement of typology with ICM simulations in this regard thus provides a compelling explanation for why we observe typological variation for some inventory sizes but not others.

5.3.4 6 Vowels

The three typologically observed 6-vowel systems 5S1 /ieaouə/, 6L0 /ieɛaou/, and 6R0 /ieaɔou/ also dominate ICM outputs, as shown in Figure 5.8.

5S1 dominates for low to moderate F2 scaling, but is overtaken by 6L0 for stronger F2 scaling. This is similar to Schwartz et al. (1997b), who predicted 5S1 for moderate F2 scaling ($\lambda > 0.25$) and 6L0 for stronger F2 scaling. For the weakest F2 scaling ($\lambda > 0.7$), Schwartz et al. (1997b)'s model also predicts the unattested 4L2b system /iæɒuyw/, which never emerges in the WL model and almost never in the RS model.

Before 6L0 overtakes 5S1, the match between ICM emergence rate and typological frequency across the three attested systems is remarkably close, with extremely high correlation for F2 scaling between 2.0 and 2.5.

Within 5S1 systems, there is some variation in output between a slightly lowered high cen-

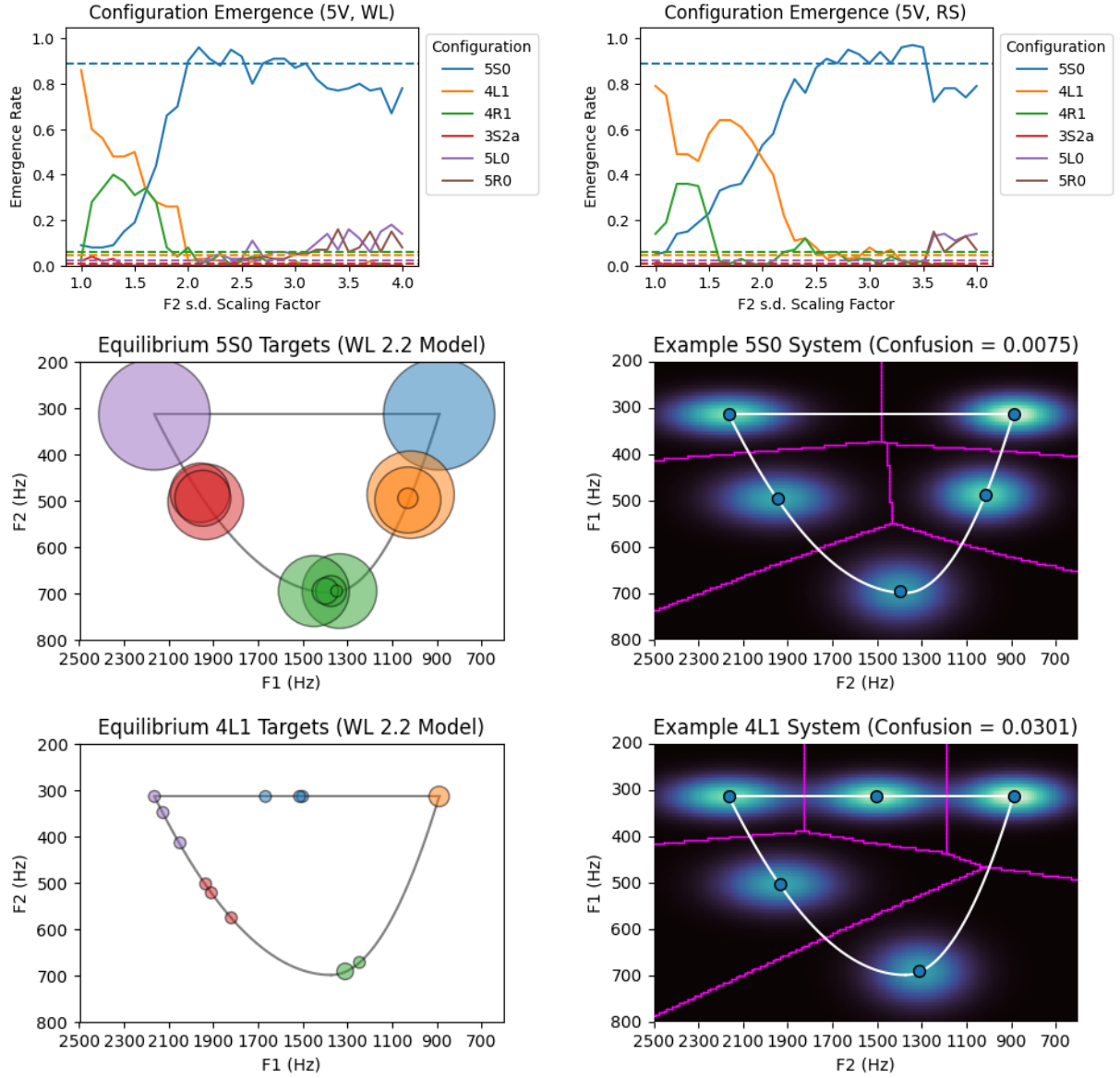


Figure 5.7: ICM results for inventory size 5.

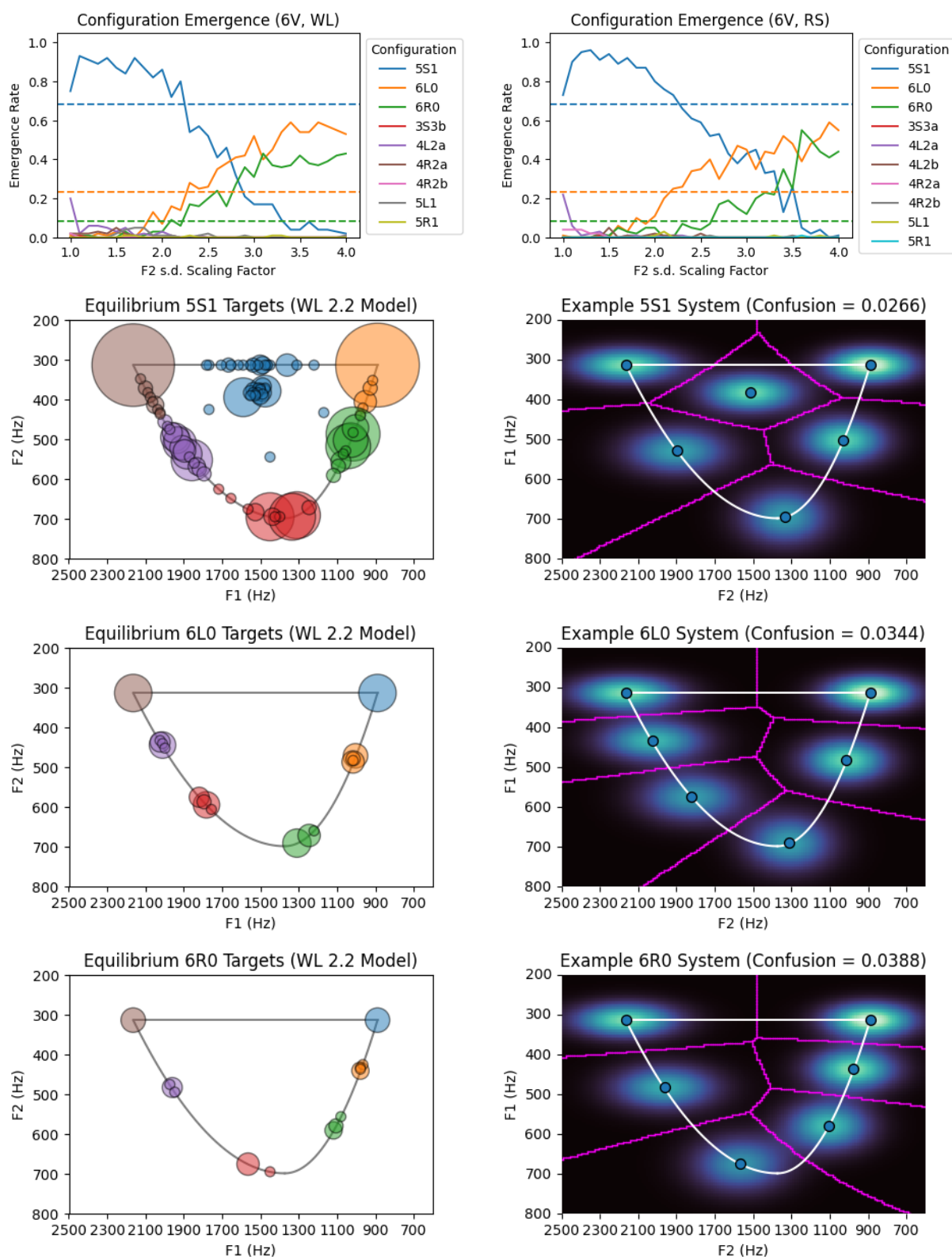


Figure 5.8: ICM results for inventory size 6.

tral vowel [ə] and a fully high central vowel [ɨ]. This actually seems to match the distinction made by Schwartz et al. (1997a) between systems with /ə/ and /ɨ/ based on transcriptions in UPSID found to be acoustically unsupported by Becker-Kristal (2010), although some experimentation not reported here suggests that these [ɨ] systems are unstable equilibria, which tend to evolve into [ə] systems if perturbed. There is also some variation among the peripheral vowels of this system, but a general pattern resembling [ieaouə] is clear.

5.3.5 7 Vowels

There are three attested 7-vowel systems: 7S0 /ieɛaɔu/ (52.3%), 5S2a /ieaouiə/ (29.5%), and 6L1 /ieɛaoui/ (18.2%). Each of these dominates ICM outputs under some parameter setting, as seen in Figure 5.9.

In contrast to previously discussed inventory sizes, there is no parameter setting that has good correlation between ICM emergence rate and typological frequency across all attested systems. The most common system, 7S0, dominates with strong F2 scaling but does not emerge at all with low F2 scaling, while the second most common system, 5S2a, has the exact opposite behavior. Meanwhile, the least common attested system 6L1 dominates for moderate F2 scaling factor between 2 and 2.5 for which other inventory sizes had the best correlation.

Again, this pattern is broadly similar to the results of Schwartz et al. (1997b), whose model predicted 7S0 with very strong F2 scaling ($\lambda < 0.22$), 6L1 with moderate F2 scaling ($\lambda < 0.4$), and the unattested 5S2b system (with the wrong arrangement of non-peripheral vowels) with low F2 scaling ($\lambda > 0.4$).

It is worth noting that the 5S2a system shows significant variation in terms of the output of its non-peripheral vowels. Some of these look more natural than others, but even for cases that look less natural, the prediction of systems with any sort of interior vowels is appealing regardless, since these are generally not predicted by models using energy functions.

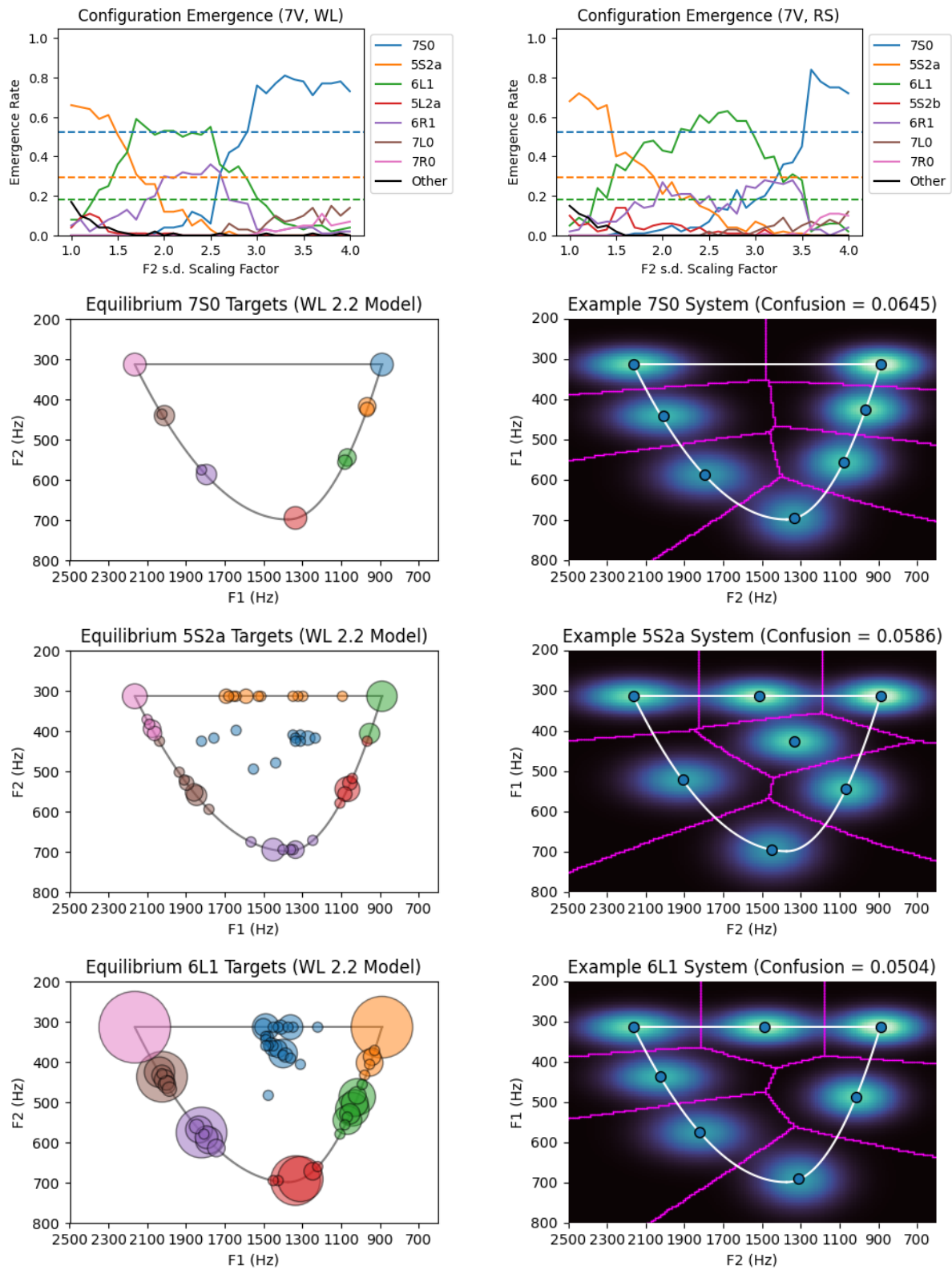


Figure 5.9: ICM results for inventory size 7.

The 6L1 systems /ieɛaoui/ shown here have relatively little variation, almost all with a fully high central vowel [i]. Becker-Kristal (2010)’s data set suggests that the non-peripheral vowel of 6L1 systems actually show significant variation, and that this vowel is typically lower than [i].⁵ In this sense, ICM predictions for 6L1 systems are somewhat dissatisfying with the current parameter setting (F2 scaling 2.2), but other parameter settings have a larger percentage of lowered non-peripheral vowels.

5.3.6 More than 7 Vowels

Figure 5.10 shows emergence rate of different systems across ICM parameter settings for inventory sizes 7 through 12.

As for seven vowels, there again does not exist any parameter setting that gives good correlation between emergence rate and typological frequency across attested systems. Although not shown, the exact formant values of predicted non-peripheral are also a bit quirky vis-à-vis acoustic typology, which we also saw with seven vowels in the 5S2a /ieaouiə/ and 6L1 /ieɛaoui/ systems.

Despite this, the ICM simulations are still extremely successful in terms of predicting the typologically dominant systems at each inventory size, predicting all typologically dominant systems under some parameter setting and a dominant system for every inventory size up to 11 using the WL model with 2.7 F2 scaling, as discussed in Section 5.1.

⁵Schwartz et al. (1997a) count 5 systems reclassified as 6L1 with /ə/ against 3 with /i/ in UPSID, further suggesting that 6L1 typically has a lowered non-peripheral vowel.

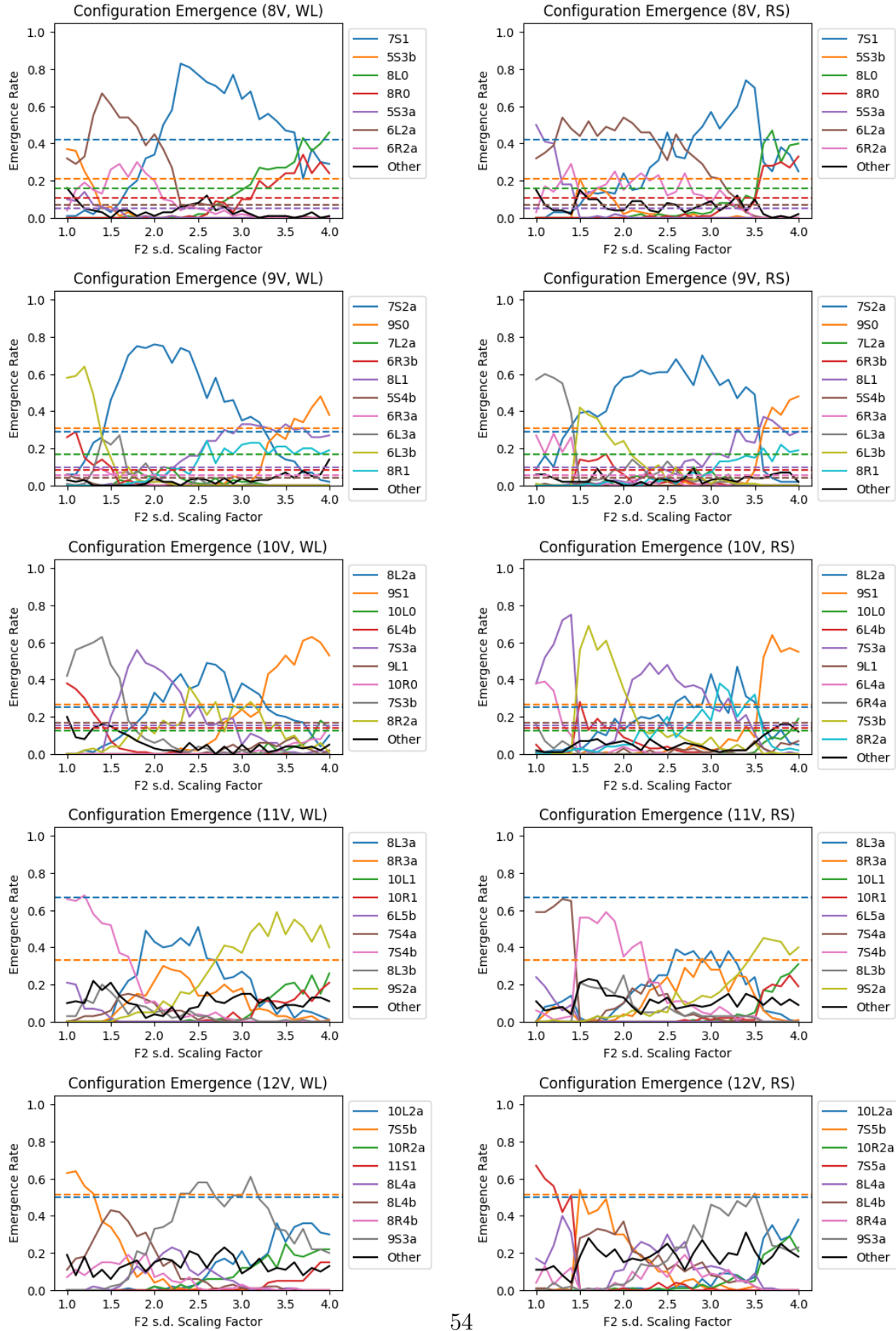


Figure 5.10: Configuration Emergence for Inventory Sizes 8 through 12.

CHAPTER 6

Discussion

The previous sections showed that ICM simulations over the vowel space meet or exceed past baselines set by Schwartz et al. (1997b) in predicting typologically dominant vowel systems and de Boer (1999) in predicting typological frequency of different vowel systems. The following discussion sections contextualize these results within past work, evaluate the level of idealization made in these simulations, and suggest directions for future research.

6.1 Tying Together Past Work

Despite its success in characterizing typologically dominant systems as perceptually optimal, Dispersion Theory has constituted by and large a theory without mechanism. On the other hand, mechanisms of emergent dispersion have either represented unrealistic models of sound change (cf. de Boer, 1999, 2000) or taken more limited scope than computer simulations accounting for vowel system typology (Blevins, 2004, p. 285-289; Boersma and Hamann, 2008). The current work presents a way of tying together these various approaches.

Although precise details may differ, the diachronic models of Blevins (2004, p. 285-289) and Boersma and Hamann (2008), which both involve restructuring of phonetic distributions in a way that reduces confusion, can be understood broadly as framework-specific implementations of Iterated Confusion Minimization. Although the goal-directed nature of Dispersion Theory has been viewed as inconsistent with the non-teleological characterization of sound change central to Blevins (2004)'s Evolutionary Phonology (cf. Vaux and Samuels, 2015),

the current work shows that the findings of Dispersion Theory emerge as a direct mechanical result of mathematically rigorous computer simulation compatible with Blevins’s own mechanism of chain shift.

The resolution of this discrepancy lies in an understanding of the relevant sound change mechanism as local optimization. Although local optimization algorithms are not guaranteed to always find globally optimal solutions (and indeed, ICM frequently does not), they may be able to find global optima most of the time, as the current work shows for ICM. Whether or not proponents of Evolutionary Phonology wish to characterize sound change as goal-directed in name, a mathematically rigorous approach to their own chain shift mechanism reveals it as goal-directed in effect.

Despite its diachronic approach, the current work shares its emphasis on optimality with synchronic approaches to phonology, currently dominated by Optimality Theory (Prince and Smolensky, 1993). However, ICM shows how optimality can arise through sound change without needing to be encoded in the synchronic grammar (cf. Flemming, 1995). By computationally modeling how realistic sound change mechanisms interact with mathematically rigorous definitions of optimality, it becomes possible to reveal optimality as an expected emergent result of sound change.

6.2 Evaluation of Idealizations

Given the results of Chapters 4 and 5, the idealizations made in the current work evidently do not impede explanation of major trends in vowel system typology. Nonetheless, it is worth taking stock of these idealizations so that future work might improve upon them.

Most of these idealizations have to do with modeling phonetic distribution shape. Although past work in Dispersion Theory has largely ignored the question of distribution shape altogether, the success of the current simulations highlights the explanatory power of phonetic distribution models informed by empirical data. Many of the current idealizations were

motivated by limited availability of such data, underscoring the importance of detailed acoustic studies of the world’s languages and transparent data reporting and sharing consistent with open science.

A future with greater availability of vowel acoustic data including means and standard deviations of higher formants and finer details such as formant covariance, skew, and kurtosis may enable more realistic modeling beyond the current idealization of independent bivariate normal distributions over F1 and F2. Some exploratory simulations not reported here suggests that such details are unlikely to significantly alter results, however. Replacing normal distributions with scaled and shifted beta distributions, which arguably better reflect real vowel distributions in having finite ranges and skew towards the center of the vowel space, yielded broadly similar results.

An apparent idealization of the current simulations relative to past dispersion models is non-use of F2'. In practice, however, it is not obvious that this makes any substantive difference in model output. The primary determiner of output vowel system configurations is the geometry of the formant space under consideration, which does not change whether we call its second dimension F2 or F2'. Although taking into account higher formants may alter the effective F2 variance with respect to human speech perception, the current work already approaches modeling of phonetic variance with higher empirical rigor than previous work.

The current assumption of fixed standard deviations as a function of vowel target is at odds with a view commonly taken in past work within Dispersion Theory that speakers should adjust phonetic variance to maintain perceptual contrast in languages with higher or lower dispersion, although empirical support for this claim has proved basically equivocal (for a review, see Recasens and Espinosa, 2009). Although future models could incorporate this sort of effect, a significant change in outcome seems unlikely given the broadly similar results using the low- and high-variance WL and RS models as well as the fact that the primary determiner of maximal attainable level of perceptual contrast, i.e. inventory size, is

held constant in each simulation.

The simulations of the current work do not take into account the competition of pressures to maximize perceptual contrast with pressures to minimize articulatory effort, although it is worth noting that evidence for this effect is mixed Disner (1983); Jongman et al. (1989); Bradlow (1995); Guion (2003); Meunier et al. (2003); Al-Tamimi and Ferragne (2005); Recasens and Espinosa (2006, 2009). Although future ICM models could be augmented to address this competition, its primary effect, namely a smaller vowel space for smaller inventories and larger space for larger inventories, can be approximated by adjusting the hard boundaries placed on the vowel space by Equations 3.1-3.3, which did not yield significantly different results in exploratory simulations.

6.3 Future Directions

Applications of Dispersion Theory have largely been limited to the vowel space (though cf. Schwartz et al., 2007), in large part because perceptual contrast between vowels can be well approximated by measures of phonetic distance over a continuous low-dimensional formant space. ICM can be simulated over any system of overlapping phonetic distributions, even those whose confusion cannot be approximated as straightforwardly with measures of phonetic distance, possibly allowing for better generalization to consonant spaces.

The current work makes the simplifying assumption of equally probable phonemes, under which the optimal listener follows maximum likelihood behavior. However, the optimal listener should actually follow Bayesian behavior, taking into account unequal prior probabilities of different phonemes and conditional probabilities of phonemes given different phonotactic contexts (cf. Kleinschmidt and Jaeger, 2015). Since phoneme frequency and phonotactics are both language-specific factors, this suggests a way to better explain language-specific deviations from dominant typological trends, a common source of criticism towards Dispersion Theory.

One potential application of Bayesian ICM is elucidating probability of phonemic merger. Preliminary ICM simulations not reported here assuming Bayesian perceptual behavior show that the perceptually optimal sound systems tend to cluster together phonemes that occur with lower probability. Insofar as one expects phonemes clustered closer together to more likely undergo phonemic merger, this predicts that phonemes occurring in fewer minimal pairs should be more likely to merge, since phonemes with lower prior probability are less likely to occur in minimal pairs due to being less likely to occur in general. This is the exact prediction of the long-standing functional load hypothesis, which Wedel et al. (2013) found to have significant empirical support in a large corpus study. Computational modeling of prior probability effects on dispersion through Bayesian ICM simulations may thus help shed further light on this topic.

Bayesian ICM may also help elucidate interesting interactions of vowel dispersion with allophony. For example, Japanese has the non-dominant 5-vowel system /ieaou/ (4L1) lacking a rounded high back vowel /u/, which has the effect of reducing perceptual contrast between /i/ and /u/ and increasing perceptual contrast between /o/ and /u/ relative to contrasts with /u/ in the globally optimal system /ieaou/. However, Japanese onset consonants show significant allophony helpful in disambiguation of /i/ vs. /u/, e.g. /si/ [çi] vs. /su/ [sɯ], but less so in disambiguation of /u/ vs. /o/, e.g. [sɯ] vs. [so]. Preliminary Bayesian ICM modeling suggests the deviation of Japanese from the typologically dominant /ieaou/ may be attributable to Japanese having a different language-specific optimal system. In particular, it is mathematically advantageous with respect to Bayesian confusion to prioritize the /u/-/o/ contrast because the /i/-/u/ contrast is often reinforced by disambiguating allophones.

Lastly, the existence of allophony itself appears as a prediction of ICM. Allophony seems to conflict with dispersion because it requires a phoneme to occupy a larger region of phonetic space, causing increased overlap with adjacent phonemes. However, preliminary ICM simulations over sequences of phonemes rather than individual phonemes (as explored in the

current work) show that allophony can reduce overall confusion between sequences under certain circumstances. For instance, suppose a language has a 3-vowel system /iau/ with low phonetic overlap and a dorsal /k/-/q/ contrast with high phonetic overlap. Even though splitting /iau/ into allophones [iæu] and [eəo] before /k/ and /q/ respectively increases overlap and hence confusability within the vowel space, doing so is mathematically advantageous for the sequences {/ik/, /ak/, /uk/, /iq/, /aq/, /uq/} because the benefit these allophones provide in disambiguating the difficult /k/-/q/ contrast outweighs the cost of increased overlap among the easily distinguished vowels. This particular pattern of allophony is widespread in 3-vowel languages like Arabic and Inuktitut.

CHAPTER 7

Conclusion

Despite the at times sharp divide between synchronic and diachronic approaches to vowel typology and dispersion, the current work presents a way to bridge the gap between these two perspectives. By recasting the confusion minimizing properties of existing diachronic models into the mathematically rigorous terms of maximum likelihood confusion, Iterated Confusion Minimization shows that the typological dominance of vowel systems with maximal perceptual contrast, successfully captured by numerous synchronic models which however lack a satisfactory explanatory mechanism, emerges as a mechanical result of a simple and realistic model of sound change. Compared with models equating perceptual contrast with psychoacoustic distance, the overlap-based approach of ICM yields not only a more nuanced understanding of perceptual contrast, but also much greater predictive power in explaining typological variation for smaller inventories and typologically dominant systems including interior vowels for larger inventories. Given the success of the current work despite a high degree of idealization, less idealized ICM models left to future work hold great promise in further elucidating aspects of the phonological typology of human language, whether in vowel systems or other aspects of sound structure.

APPENDIX A

Scale-Invariance of Maximum Likelihood Confusion

Whereas distance-based measures of perceptual contrast are highly sensitive to choice of acoustic scale, maximum likelihood confusion for a given set of phonetic distributions is invariant with respect to scales that are monotonic transformations of one another.

To illustrate this, consider Figure A.1. On the left we have a system of three phonemes with overlapping distributions shown along a linear scale of some phonetic value. To the right, we see the same exact set of distributions instead measured in a logarithmic scale. The intersection points of the three probability density functions and hence the optimal ML boundaries in the logarithmic plot to the right are simply the logarithmically scaled values of the ML boundaries in the linear plot to the left, i.e. $\ln(6) = 1.79$ and $\ln(8) = 2.08$.

Despite the change in scale, the degree of overlap in the plots of Figure A.1 turns out the same. Even though the region of overlap between Phonemes 1 and 2 has the same width as the region of overlap between Phonemes 2 and 3 in the linear scale but not the logarithmic scale, and even though both regions of overlap have a much larger width in the linear scale than the logarithmic scale, this difference in width is balanced by counteracting changes to the height of the regions of overlap reflecting changes in probability density with respect to scale such that the areas of overlap remain the same.

This makes sense given the monotonicity of the logarithmic function. All tokens to the left of the boundaries at 6 and 8 in the linear plot will also be to the left of the corresponding boundaries at $\ln(6)$ and $\ln(8)$ once re-scaled in the logarithmic plot (and similarly for tokens to the right of these boundaries). Thus, the same percentage of tokens from each phoneme

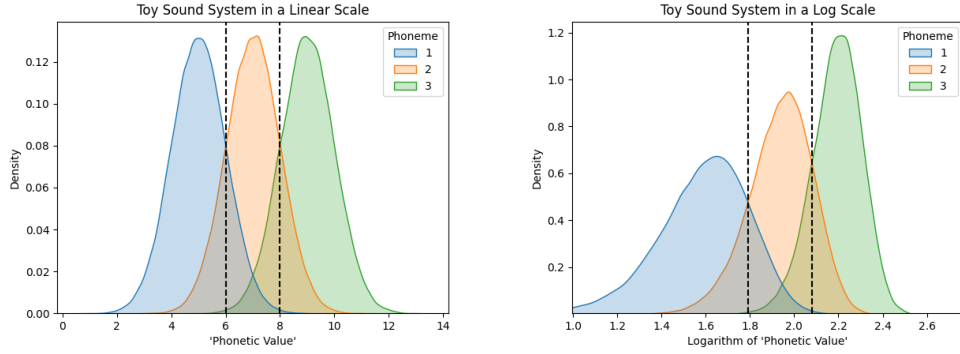


Figure A.1: Density plots for three toy phonemes each based on 100,000 tokens sampled from a normal distribution in a linear scale (left) and plots for those same tokens measured in a logarithmic scale (right). ML perceptual boundaries lie at the intersection points, which in the log scale are simply the logarithm of the boundaries in the linear scale. Since these boundaries are at the same quantile of each distribution regardless of scale, the confusion rate, i.e. percentage of tokens on the ‘wrong’ side of these boundaries, does not change.

distribution will be on the ‘wrong’ side of the perceptual boundaries using either scale, giving an overall confusion rate of 0.21 in both scales that is reflected by equal areas of overlap.¹

Generalizing this logic, the confusion rate with respect to any arbitrary perceptual boundary (including sub-optimal non-ML boundaries) for any set of distributions in these two scales is invariant. From this we can determine that the optimal boundary in the logarithmic scale always has to be a scaled version of the optimal boundary in the linear scale. Otherwise, there would need to exist some boundary with a better confusion rate in one scale than the other, which is a contradiction. Consequently, the confusion rate of a maximum likelihood classifier is invariant over conversions between linear and logarithmic scales, even as the relative distances between categories may change, e.g. Phoneme 2 is equidistant between the other phonemes in the linear scale, but closer to Phoneme 3 in the logarithmic scale.

¹Confusion in Figure A.1 is the area of overlap in either plot, with the area of triple overlap double counted (taking into account misperceived tokens of both Phoneme 1 and 3). In general, areas of overlap involving n categories must be counted $n - 1$ times towards the confusion rate.

By this same reasoning, the invariance of ML confusion holds across any monotonic rescaling, including between common frequency scales like Hz, Mel, and Bark. Hence, if we are using ML confusion as an evaluation metric over vowel systems, we have no a priori reason to prefer a psychoacoustic scale. Rather, preference of one scale over another is driven by ease with which phonetic distribution shapes can be captured. Since acoustic studies of vowel systems such as those that inform the models of Section 3.2 nearly always report values in Hz but only seldom in other scales, the Hz scale is the clear choice for the current work.

APPENDIX B

Conversion between Taxonomic Systems

The taxonomy of Schwartz et al. (1997a), henceforth S97, classifies vowel system configurations reported in UPSID using the following convention:

$$(X...Y)^{(h)}\pi_{p(a)} \quad (\text{B.1})$$

where parentheses denote an optional component of the configuration name, $X...Y$ is a list of the non-peripheral vowels in the system, h denotes a system with a ‘hole’ lacking /i/ or /u/ (e.g. /i a o/ is ${}^h\pi_3$), p denotes the number of peripheral vowels, and a is given the value l, r, b, or u for systems with left, right, bottom, or upper crowding. Note that vowels are only considered peripheral if along the front, bottom, or back periphery of the space, whereas vowels along the high periphery like /i/ or /y/ are considered non-peripheral.

Generally, S97 configurations can straightforwardly be recast into configurations from Becker-Kristal (2010)’s taxonomy, henceforth BK10, since the latter for the most part shares a subset of the distinctions of the former. Given an S97 system of the form $(X...Y)^{(h)}\pi_{p(a)}$, the BK10 system is generally $\#_i Z \#_j$, where:

- $\#_i = p$,
- Z is L if $a = l$, R if $a = r$, and S if a is not present,
- $\#_j = \text{length}(X...Y)$, and
- if $\text{length}(X...Y) > 1$, the configuration label is suffixed with b if there are multiple high non-peripheral vowels in $X...Y$ and suffixed with a otherwise.

For example, ${}^h\pi_{4l}$ becomes 4L0, $\text{ia } \pi_7$ becomes 7S2a, and both $\text{a } \pi_5$ and $\text{i } \pi_5$ become 5S1. Vis-à-vis S97, BK10 has the effect of ignoring ‘holes’ at the top corners of the vowel space and collapsing a number of distinctions between non-peripheral vowels.

Although generally straightforward following the correspondences above, reclassification is non-trivial for up- and bottom-crowded inventories S97, since BK10 only considers left- and right-crowding.

The only case of an up-crowded primary vowel system in S97 is $\text{a } \pi_2$ /iəu/. Since BK10 defines the periphery in relative rather than absolute terms, /ə/ is considered to be at the low periphery within this system, allowing a straightforward classification as 3S0. Similarly, since /e/ is the lowest vowel of π_{3l} /ieu/ and left- and right-crowding are defined relative to the lowest vowel in BK10 rather than based on the absolute number of front and back vowels as in S97, this system is also classified as 3S0 rather than 3L0 (which would imply a system like /iea/), making it the only system in S97 where l or r does not correspond to a BK10 L or R as per the correspondences above. These two reclassifications (affecting only 1 S97 language each) result in 3S0 having a 100% baseline typological frequency among 3-vowel inventories.

Bottom-crowding in S97 comes from cases of vowels with an even number of peripheral vowels including two low vowels, e.g. π_{6b} /ieæəou/. In practice, however, one of /æɑ/ will always be acoustically lower (i.e. have higher F1) than the other, so BK10 always classifies such systems as 6L0 or 6R0. For the purposes of reclassifying S97 into BK10, the current work splits the typological frequency of systems like π_{6b} between 6L0 and 6R0 based on the rate that those systems otherwise occur.

For example, S97 reports 12 π_{6l} systems and 4 π_{6r} systems, so the current work allocates the 3 reported π_{6b} systems to 6L0 and 6R0 using this same 2:1 ratio, giving a total of 14 6L0 systems and 5 6R0 systems. In the case of 11 vowel systems, this resulted in a fractional language count for 8L3a and 8R3a. In cases where there was no analogous left- or right-crowded system reported by S97, which only ever occurred for large configurations only

attested in a single language, the system was assumed to be left-crowded, since left-crowded systems are generally much more common than right-crowded systems.

Bibliography

- Adank, P., Smits, R., Van Hout, R., 2004. A comparison of vowel normalization procedures for language variation research. *The Journal of the acoustical society of America* 116, 3099–3107.
- Al-Tamimi, J.E., Ferragne, E., 2005. Does vowel space size depend on language vowel inventories? evidence from two arabic dialects and french., in: *Interspeech, CiteSeer*. pp. 2465–2468.
- Becker-Kristal, R., 2010. Acoustic typology of vowel inventories and Dispersion Theory: Insights from a large cross-linguistic corpus. University of California, Los Angeles.
- Blevins, J., 2004. *Evolutionary phonology: The emergence of sound patterns*. Cambridge University Press.
- de Boer, B., 1999. Self organization in vowel systems. Ph.D. thesis. Ph. D. Dissertation, AI-Laboratory Vrije Universiteit Brussel.
- de Boer, B., 2000. Self-organization in vowel systems. *Journal of phonetics* 28, 441–465.
- Boersma, P., Hamann, S., 2008. The evolution of auditory dispersion in bidirectional constraint grammars. *Phonology* 25, 217–270.
- Boersma, P., Hayes, B., 2001. Empirical tests of the gradual learning algorithm. *Linguistic inquiry* 32, 45–86.
- Boersma, P., et al., 1997. How we learn variation, optionality, and probability, in: *Proceedings of the Institute of Phonetic Sciences of the University of Amsterdam, Amsterdam*. pp. 43–58.
- ten Bosch, L.F.M., 1991. On the structure of vowel systems: aspects of an extended vowel model using effort and contrast. Ph.D. thesis. Uitgever niet vastgesteld.

- Bradlow, A.R., 1995. A comparative acoustic study of english and spanish vowels. *The Journal of the Acoustical Society of America* 97, 1916–1924.
- Carlson, R., Fant, G., Granström, B., 1974. Two-formant models, pitch, and vowel perception. *Acta Acustica united with Acustica* 31, 360–362.
- Carlson, R., Granström, B., Fant, G., 1970. Some studies concerning perception of isolated vowels. *Speech Transmission Laboratory Quarterly Progress and Status Report* 11, 19–35.
- Clarke, C.M., Garrett, M.F., 2004. Rapid adaptation to foreign-accented english. *The Journal of the Acoustical Society of America* 116, 3647–3658.
- Crothers, J., 1978. Typology and universals of vowel systems in phonology. *Universals of human language* 2, 95–152.
- Disner, S.F., 1983. Vowel quality: The relation between universal and language specific factors. *UCLA WPP* .
- Fisher, R.A., 1922. On the mathematical foundations of theoretical statistics. *Philosophical transactions of the Royal Society of London. Series A, containing papers of a mathematical or physical character* 222, 309–368.
- Flemming, E., 1995. Audio representations in phonology. Ph.D. thesis. Ph. D. Dissertation, University of California, Los Angeles.
- Flemming, E., 2005. A phonetically-based model of phonological vowel reduction. Manuscript. Cambridge: MIT .
- Gelman, A., Carlin, J.B., Stern, H.S., Rubin, D.B., 1995. Bayesian data analysis. Chapman and Hall/CRC.
- Grandgent, C.H., 1927. From Latin to Italian: An Historical outline of the phonology and morphology of the Italian language. Harvard University Press.

- Guion, S.G., 2003. The vowel systems of quichua-spanish bilinguals: Age of acquisition effects on the mutual influence of the first and second languages. *Phonetica* 60, 98–128.
- Jongman, A., Fourakis, M., Sereno, J.A., 1989. The acoustic vowel space of modern greek and german. *Language and speech* 32, 221–248.
- Kleinschmidt, D.F., Jaeger, T.F., 2015. Robust speech perception: recognize the familiar, generalize to the similar, and adapt to the novel. *Psychological review* 122, 148.
- Kullback, S., Leibler, R.A., 1951. On information and sufficiency. *The annals of mathematical statistics* 22, 79–86.
- Labov, W., 1994. *Principles of Linguistic Change, Volume 1: Internal Factors*. volume 1. Blackwell.
- Labov, W., Ash, S., Boberg, C., 2006. *The atlas of North American English: Phonetics, phonology and sound change*. Mouton de Gruyter.
- Liljencrants, J., Lindblom, B., 1972. Numerical simulation of vowel quality systems: The role of perceptual contrast. *Language* , 839–862.
- Lindblom, B., 1986. Phonetic universals in vowel systems. *Experimental phonology* , 13–44.
- Maddieson, I., 1984. *Pattern of Sounds*. Cambridge University Press, Cambridge, UK.
- Magnuson, J.S., Nusbaum, H.C., 2007. Acoustic differences, listener expectations, and the perceptual accommodation of talker variability. *Journal of Experimental Psychology: Human perception and performance* 33, 391.
- Meunier, C., Frenck-Mestre, C., Lelekov-Boissard, T., Le Besnerais, M., 2003. Production and perception of vowels: does the density of the system play a role?, in: *Proceedings of International Congress of Phonetic Sciences (ICPhS)*, Université Autonome de Barcelone. pp. 723–726.

- Pierrehumbert, J., 2001. Exemplar dynamics: Word frequency, lenition, and contrast. *Frequency and the Emergence of Linguistic Structure*/John Benjamins .
- Prince, A., Smolensky, P., 1993. *Optimality theory: Constraint interaction in generative grammar*. University, New Brunswick, and University of Colorado .
- Recasens, D., Espinosa, A., 2006. Dispersion and variability of catalan vowels. *Speech communication* 48, 645–666.
- Recasens, D., Espinosa, A., 2009. Dispersion and variability in catalan five and six peripheral vowel systems. *Speech Communication* 51, 240–258.
- Schwartz, J.L., Boë, L.J., Abry, C., 2007. Linking dispersion-focalization theory and the maximum utilization of the available distinctive features principle in a perception-for-action-control theory. *Experimental approaches to phonology* , 104–124.
- Schwartz, J.L., Boë, L.J., Vallée, N., Abry, C., 1997a. Major trends in vowel system inventories. *Journal of phonetics* 25, 233–253.
- Schwartz, J.L., Boë, L.J., Vallée, N., Abry, C., 1997b. The dispersion-focalization theory of vowel systems. *Journal of phonetics* 25, 255–286.
- Sedlak, P., 1969. Typological considerations of vowel quality systems. *Working Papers on Language Universals* 1, 1–40.
- Vaux, B., Samuels, B., 2015. Explaining vowel systems: dispersion theory vs natural selection. *The Linguistic Review* 32, 573–599.
- Wedel, A., Kaplan, A., Jackson, S., 2013. High functional load inhibits phonological contrast loss: A corpus study. *Cognition* 128, 179–186.

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